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Sakai et al.

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(54) **HEAT EXCHANGE UNIT**

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F24F 1/56 (2011.01)
F28D 21/00 (2006.01)

(52) **U.S. Cl.**
CPC **F28F 9/007** (2013.01); **F24F 1/56** (2013.01); **F28D 21/00** (2013.01); **F28D 2021/0068** (2013.01)

(58) **Field of Classification Search**

CPC .. **F28F 9/007**; **F24F 1/56**; **F28D 21/00**; **F28D 2021/0068**

(Continued)

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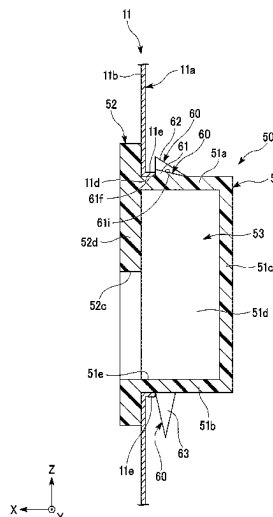
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(57) **ABSTRACT**

An aspect of a heat exchange unit of a refrigeration cycle device, includes a housing having openings, and gripping members attached to the openings. Each gripping member has a main body having an inner space that opens to an outside, and a plurality of claws that protrude from the main body and that hook to a circumferential edge of the openings. The plurality of claws include a plurality of first claw portions, and at least one second claw portion that is provided on an opposite side to the plurality of first claw portions in a predetermined direction that intersects the opening direction. At least one claw portion out of the plurality of first claw portions and the at least one second claw portion is an elastic claw that is cantilevered by the main body and is elastically deformable in the predetermined direction with respect to the main body.

10 Claims, 15 Drawing Sheets



(58) **Field of Classification Search**

USPC 165/165

See application file for complete search history.

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FIG. 1

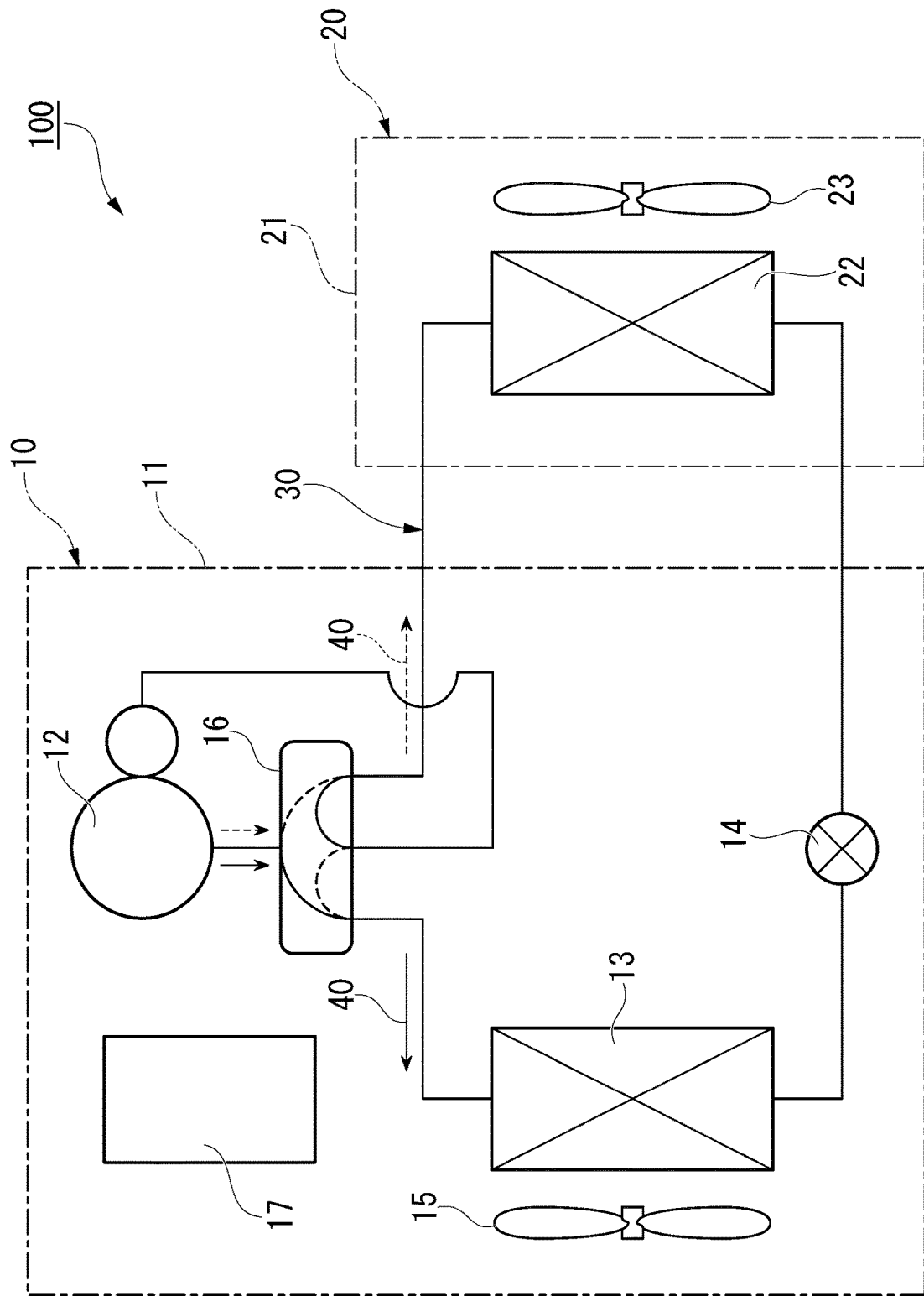


FIG. 2

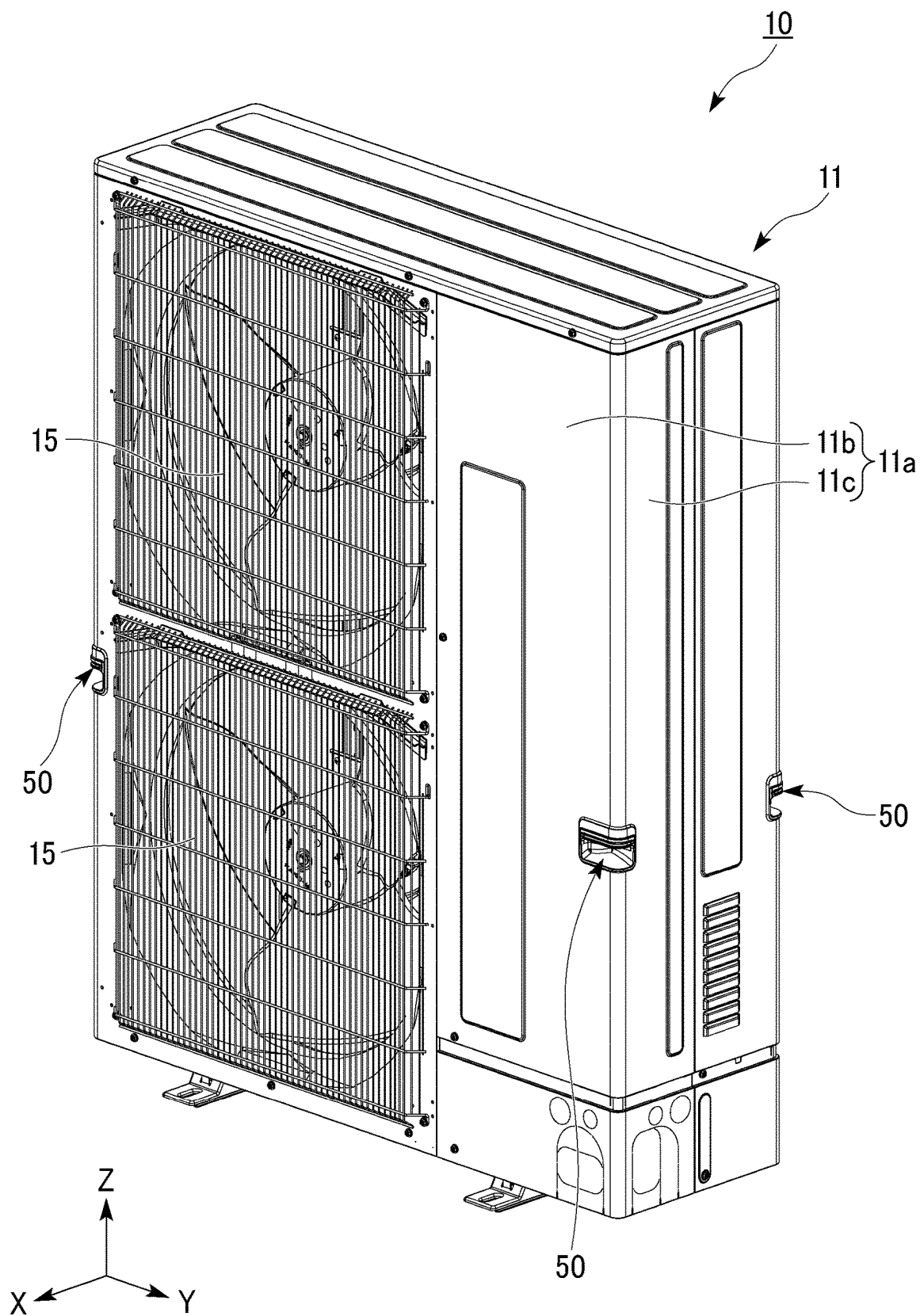


FIG. 3

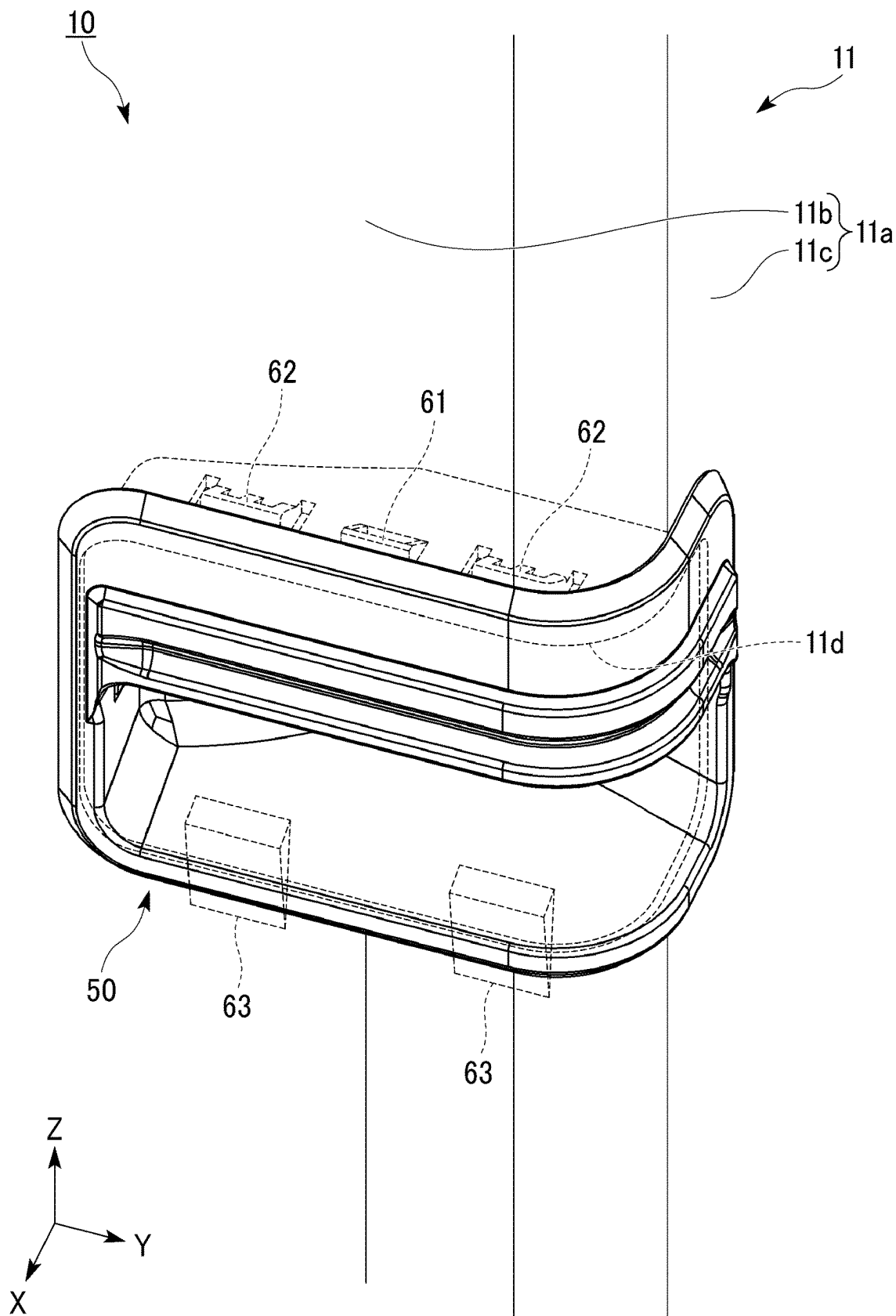


FIG. 4

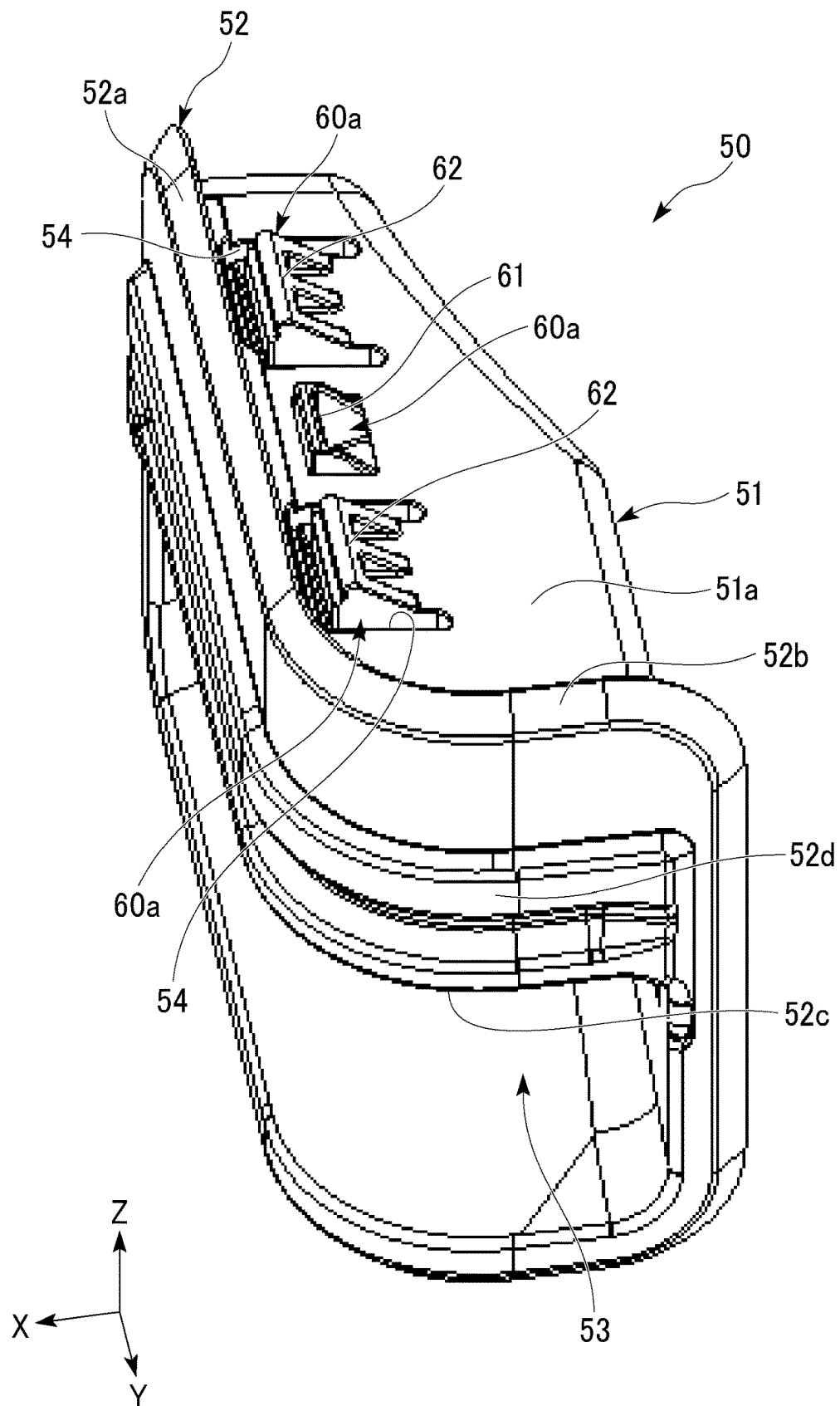
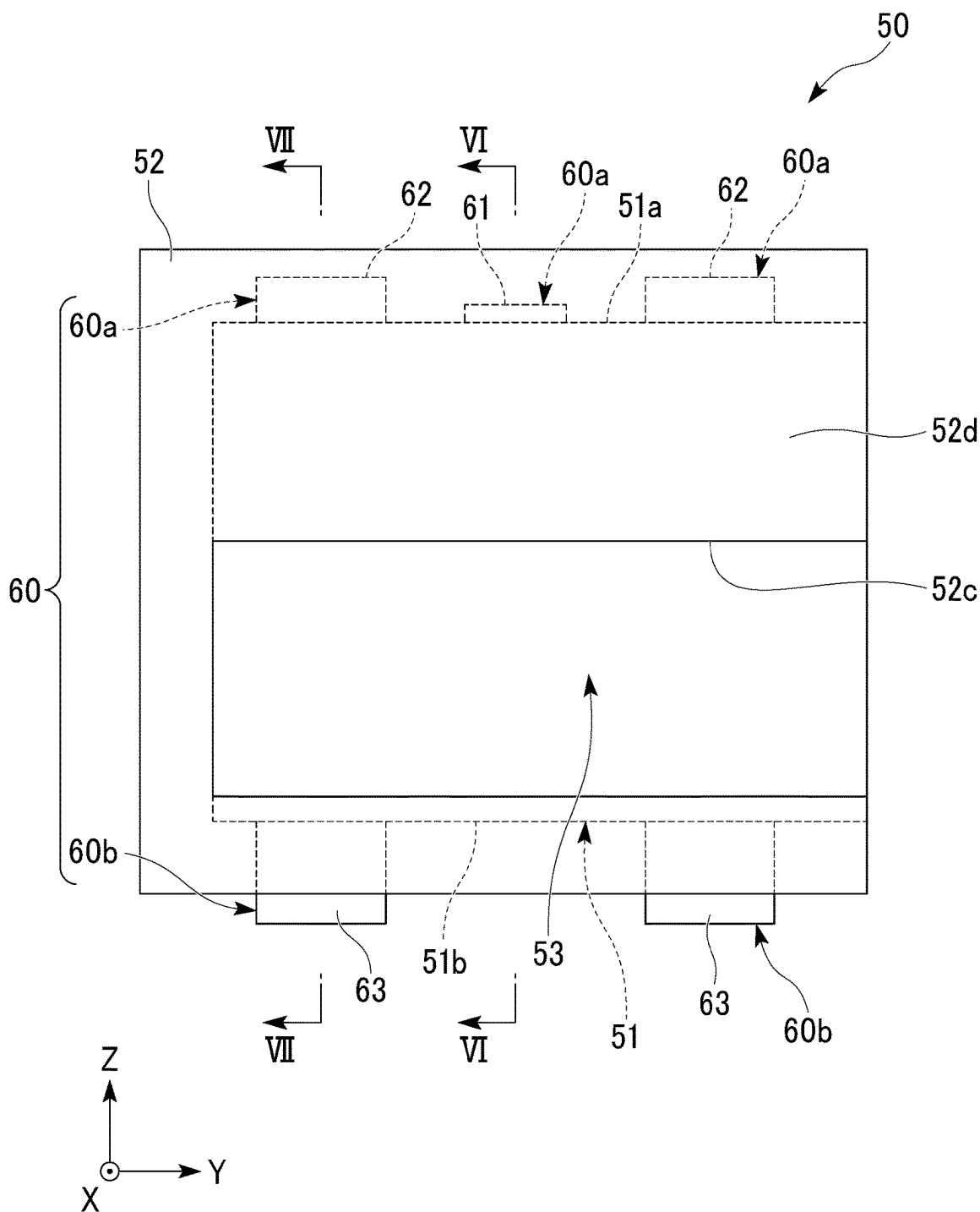
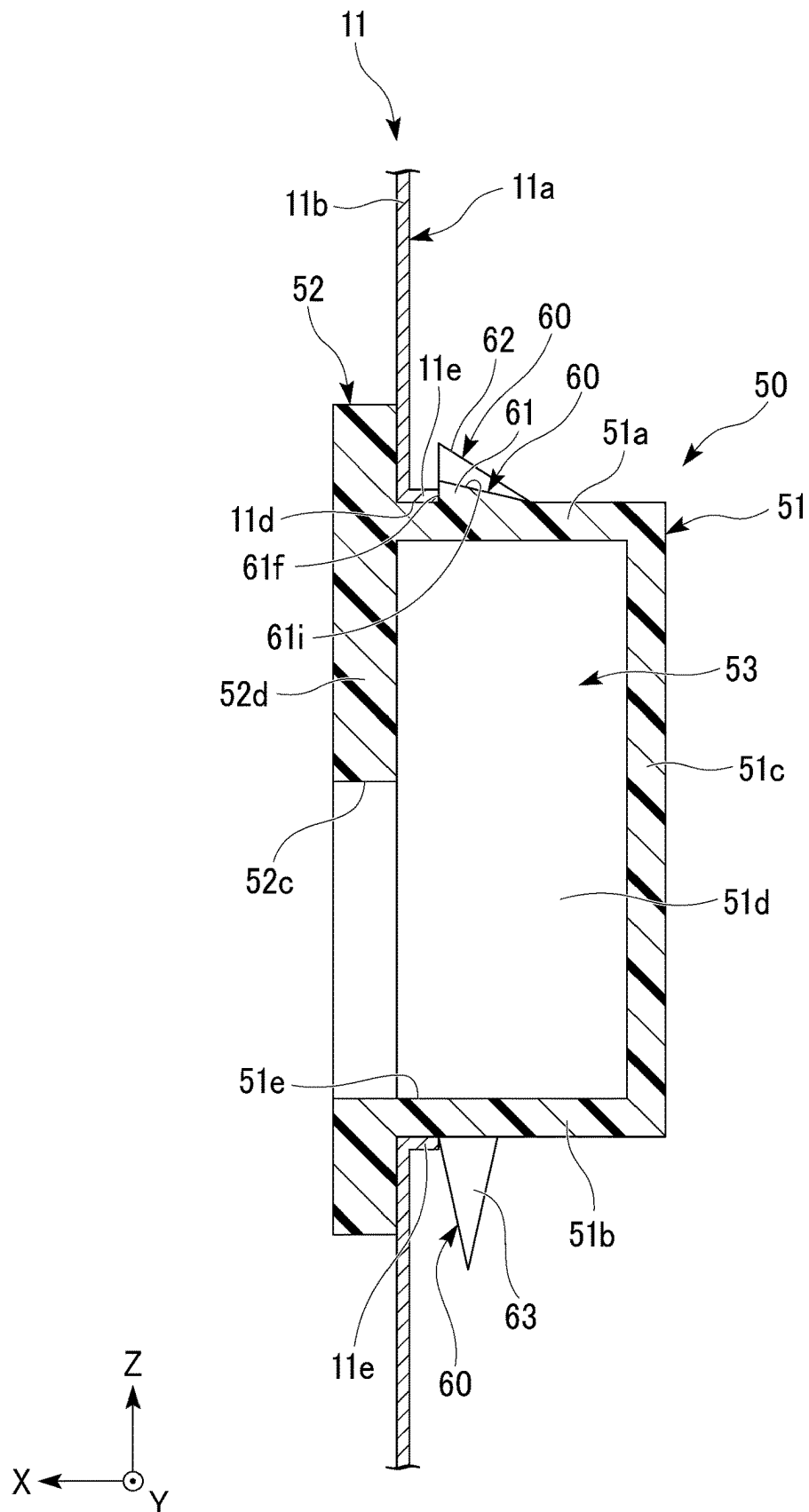


FIG. 5





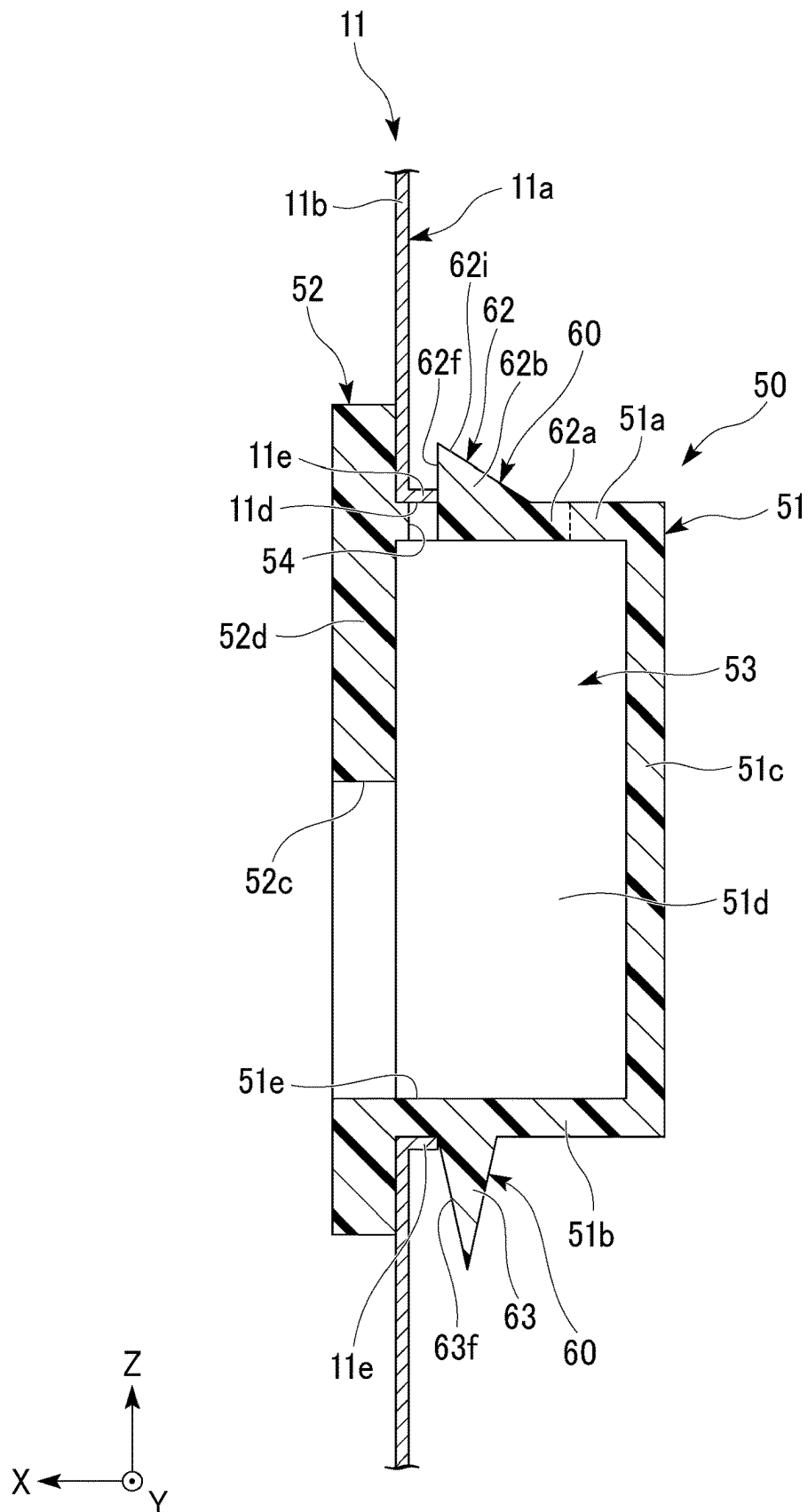


FIG. 8

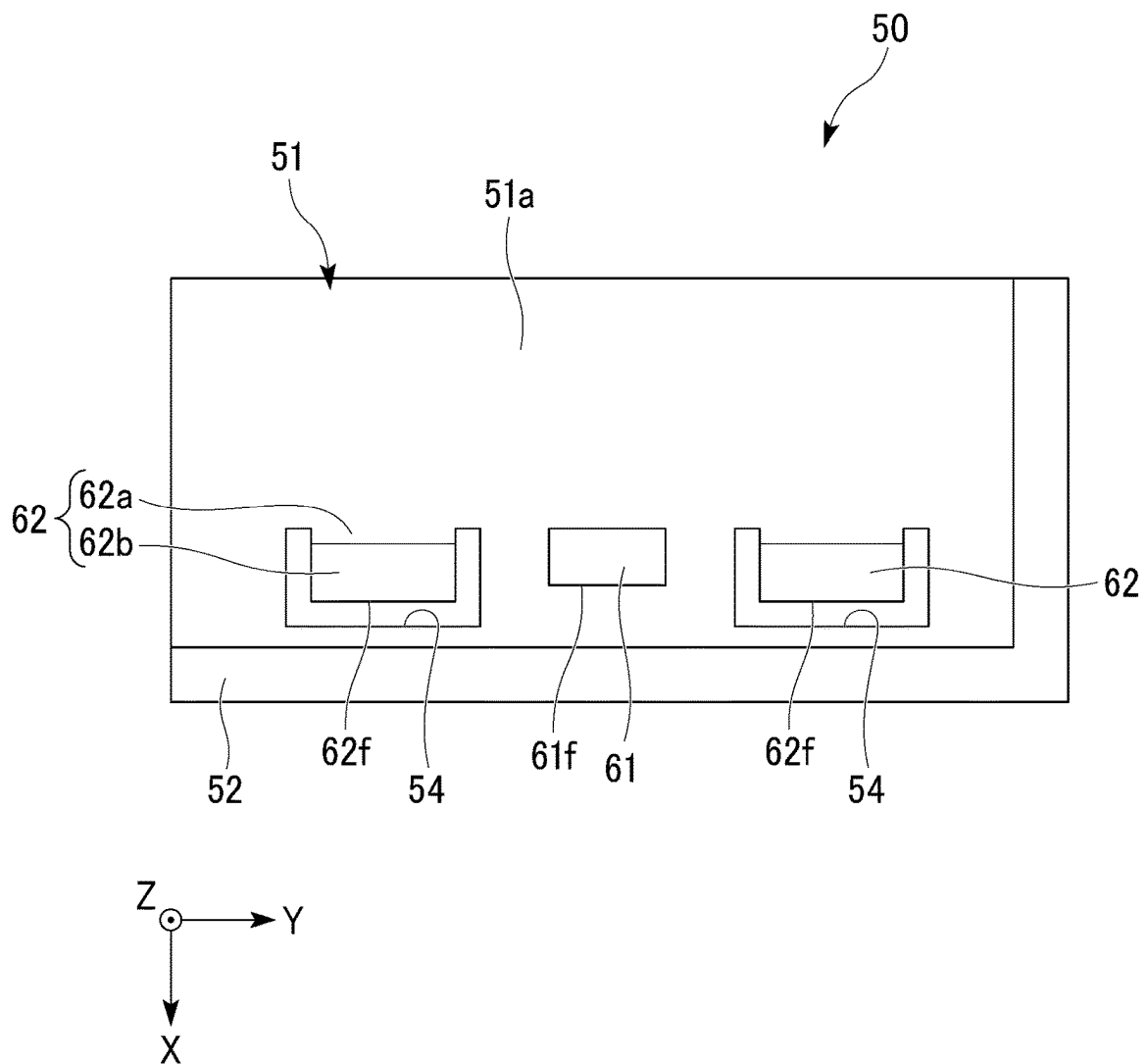


FIG. 9

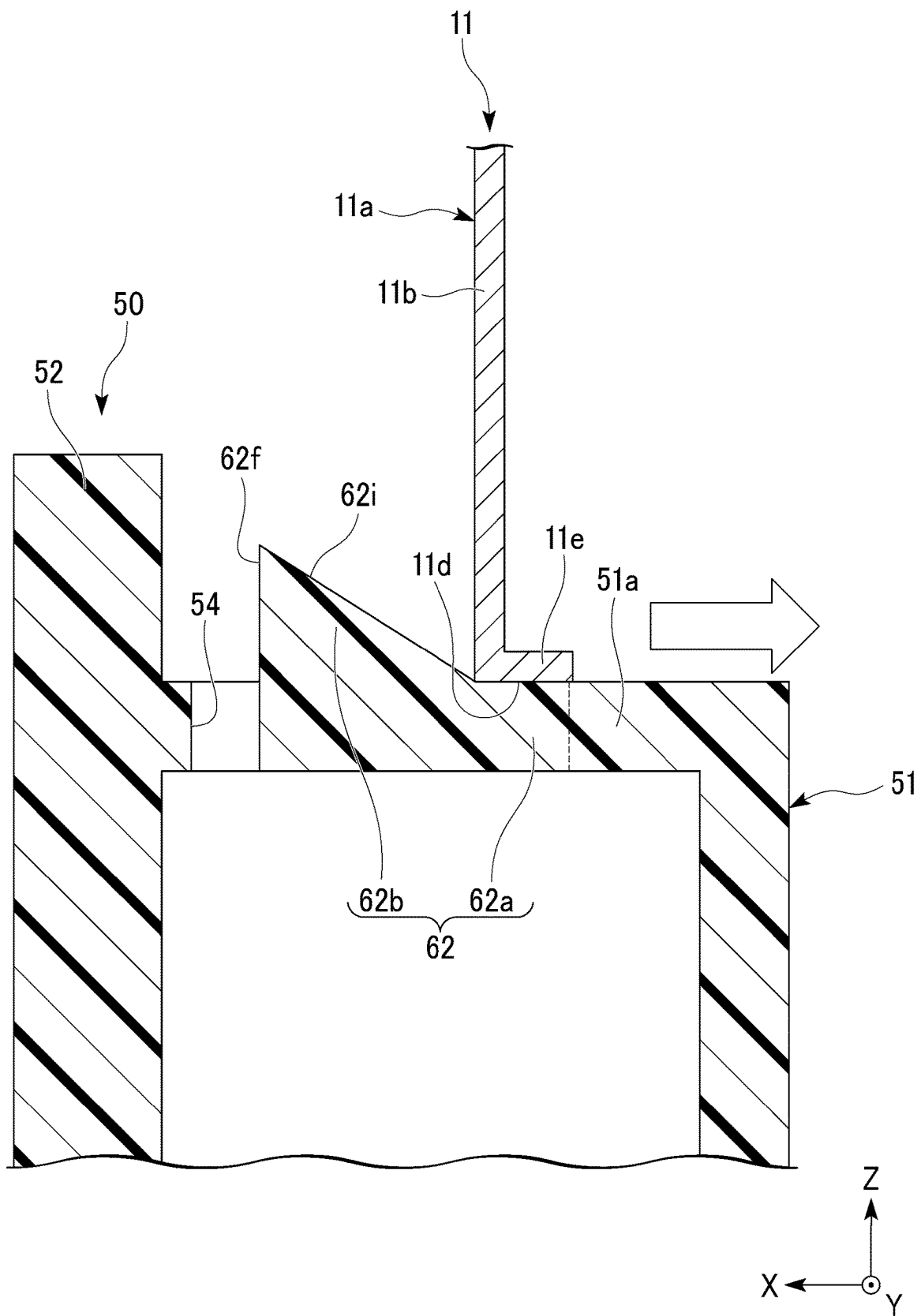


FIG. 10

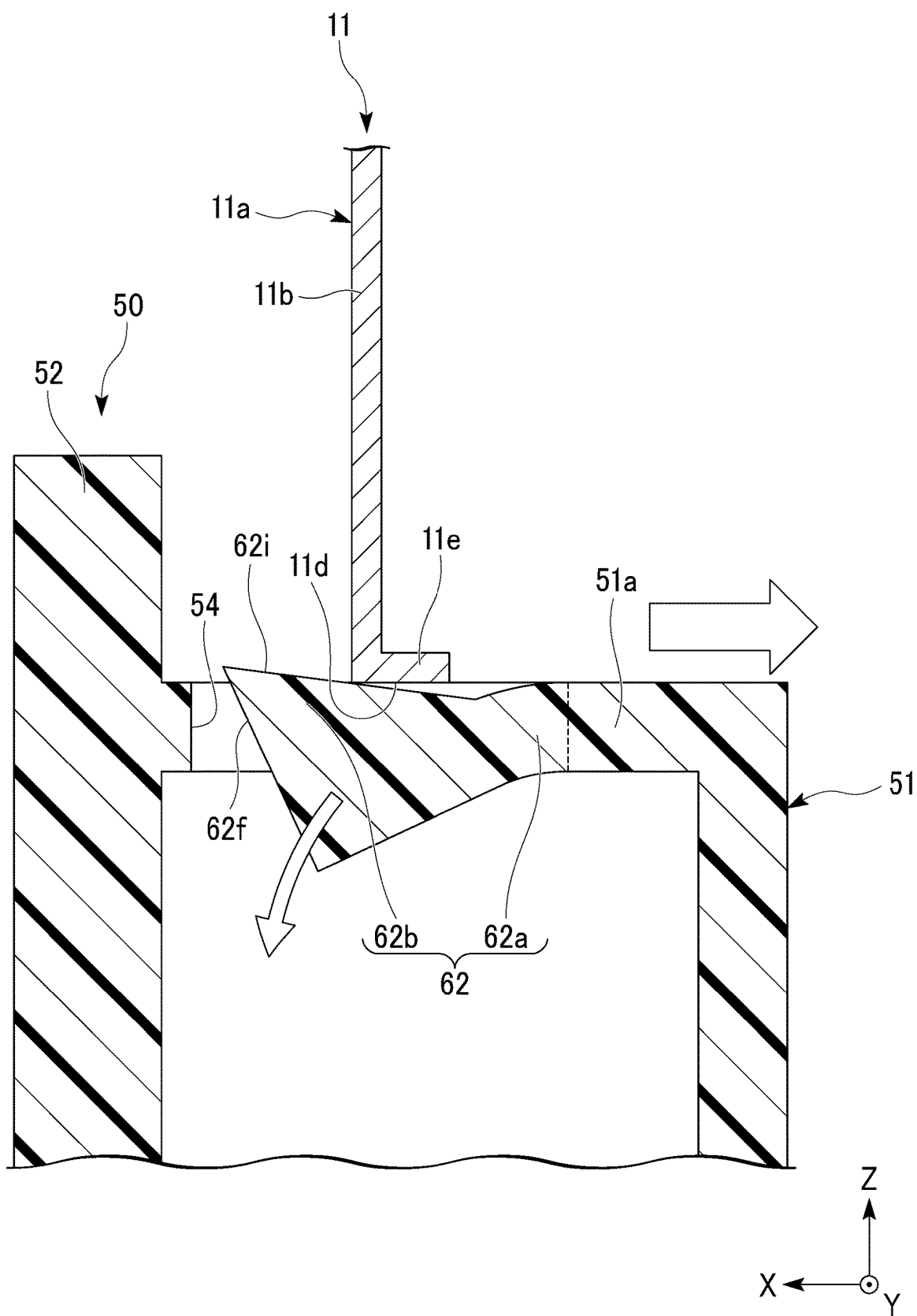


FIG. 11

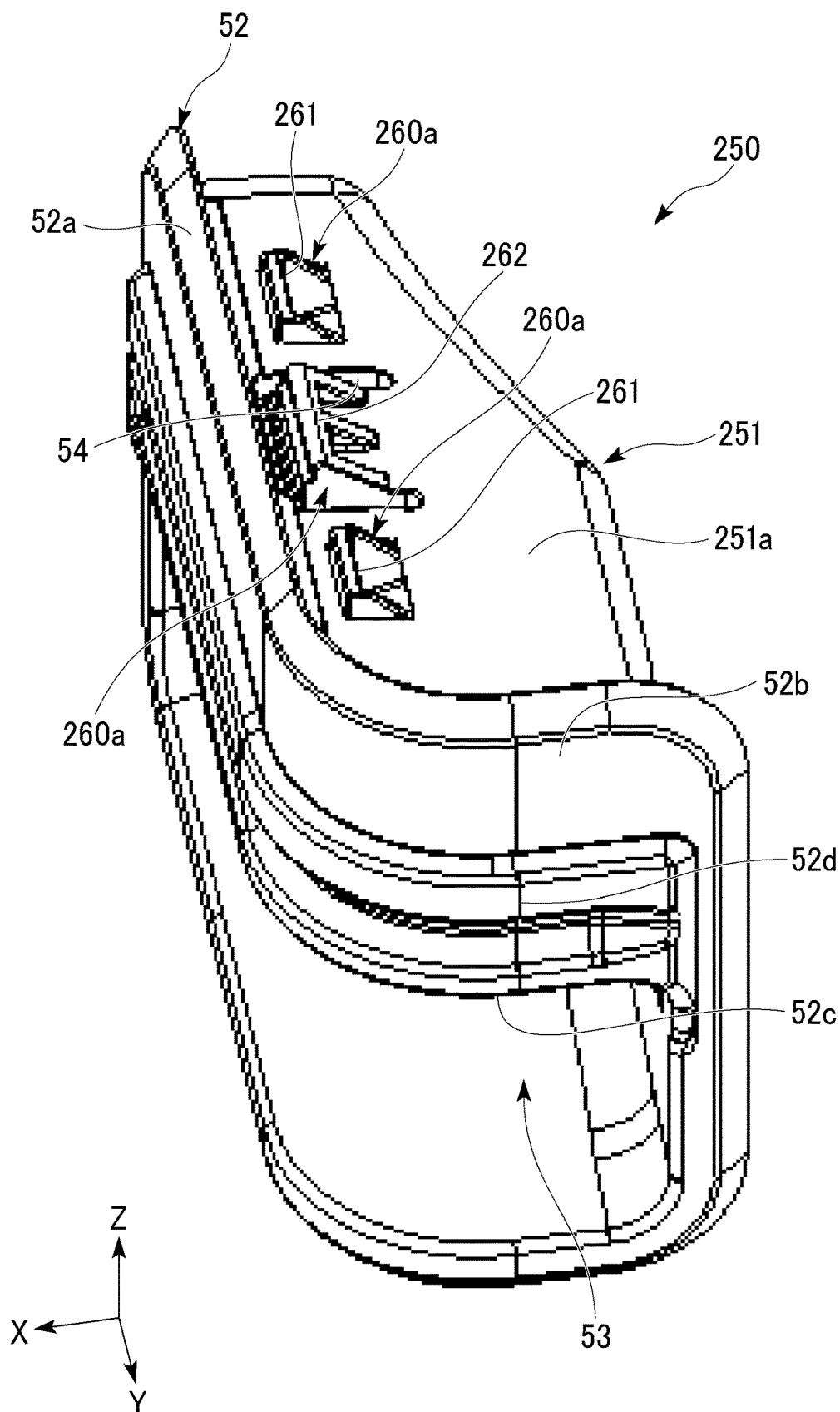


FIG. 12

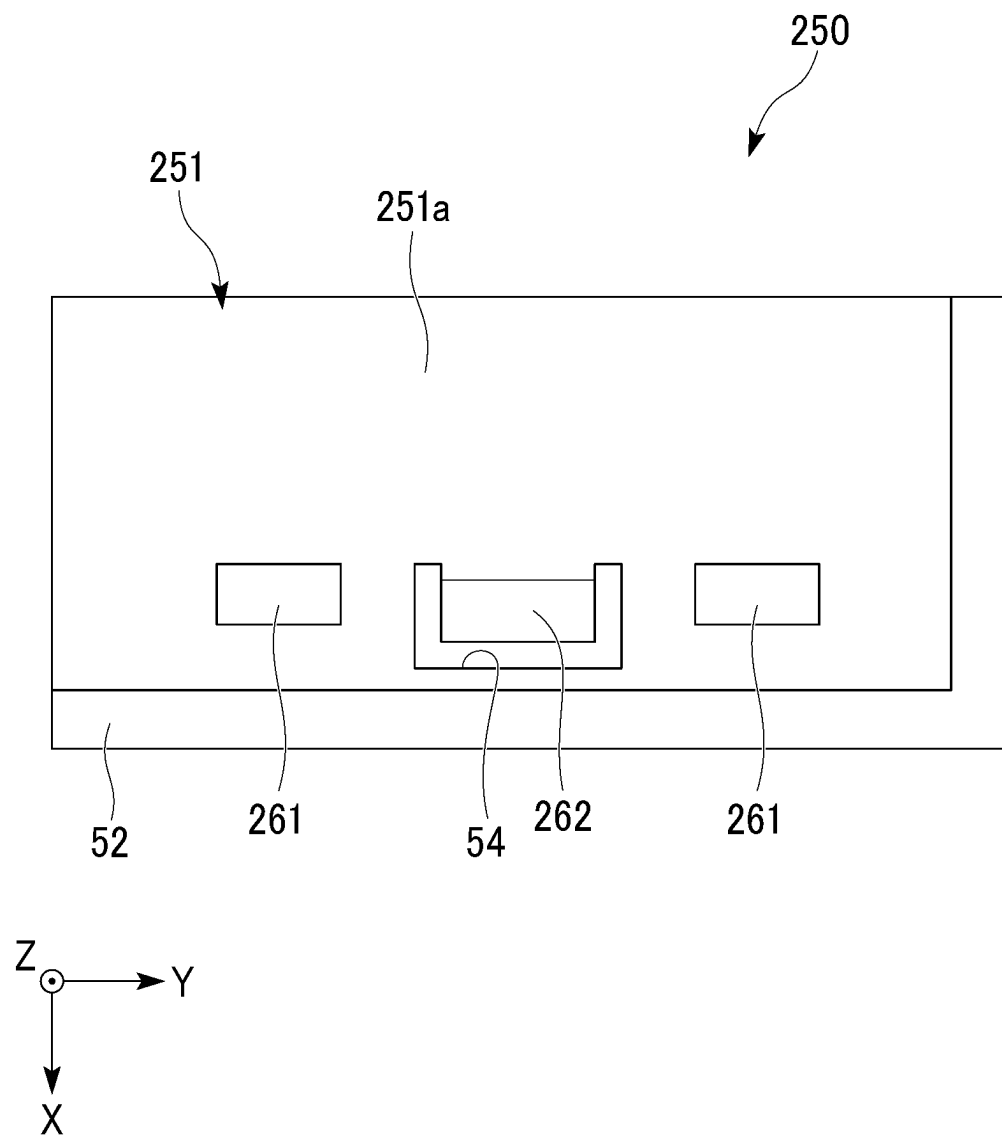


FIG. 13

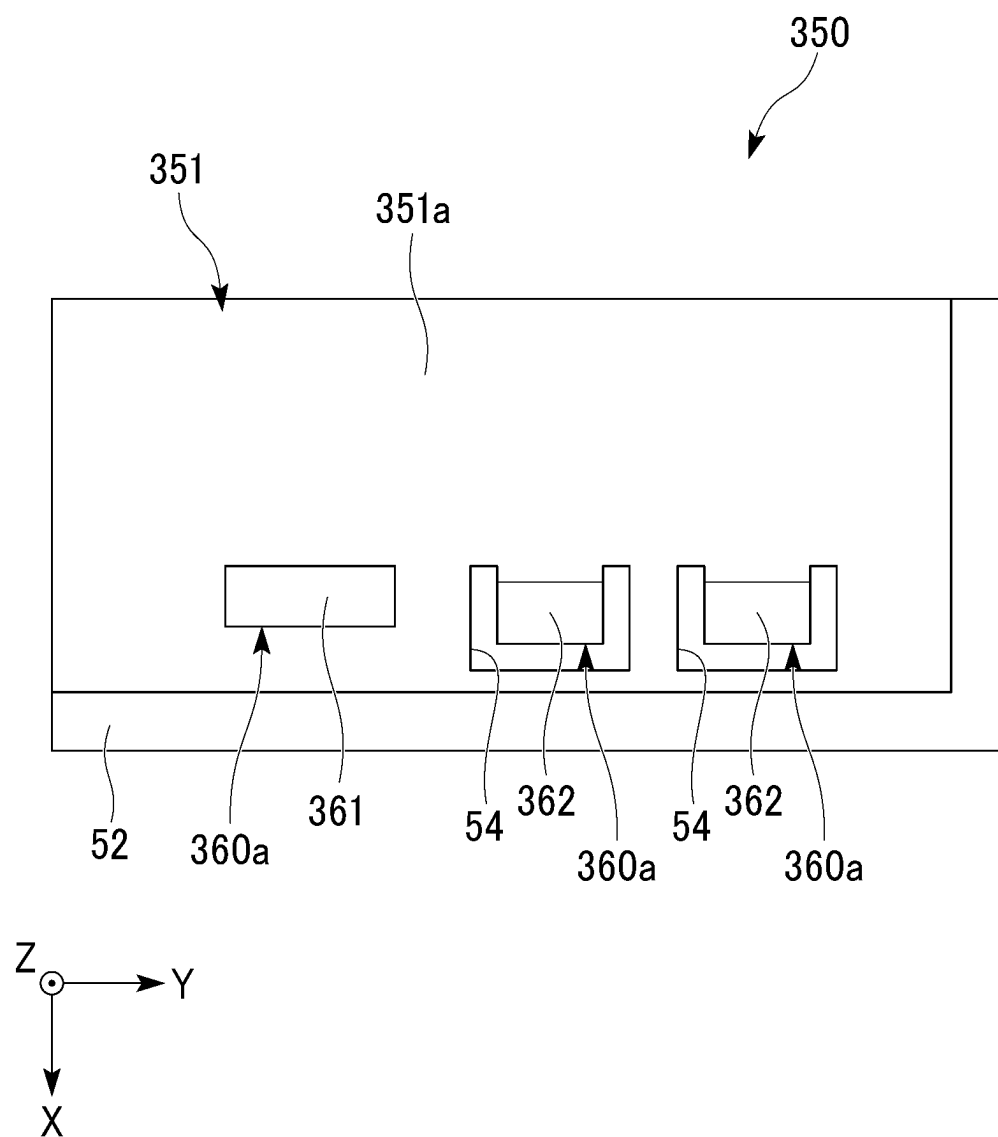


FIG. 14

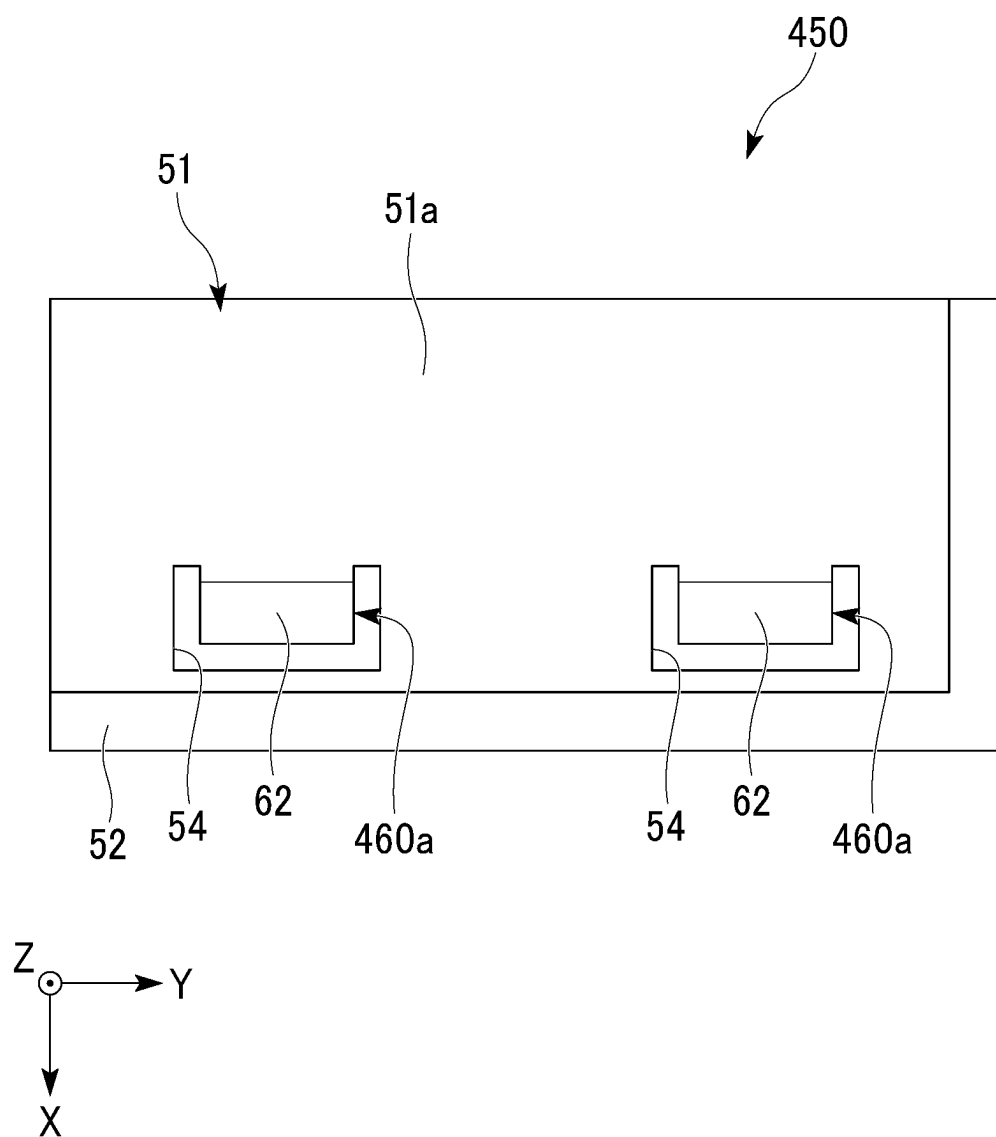
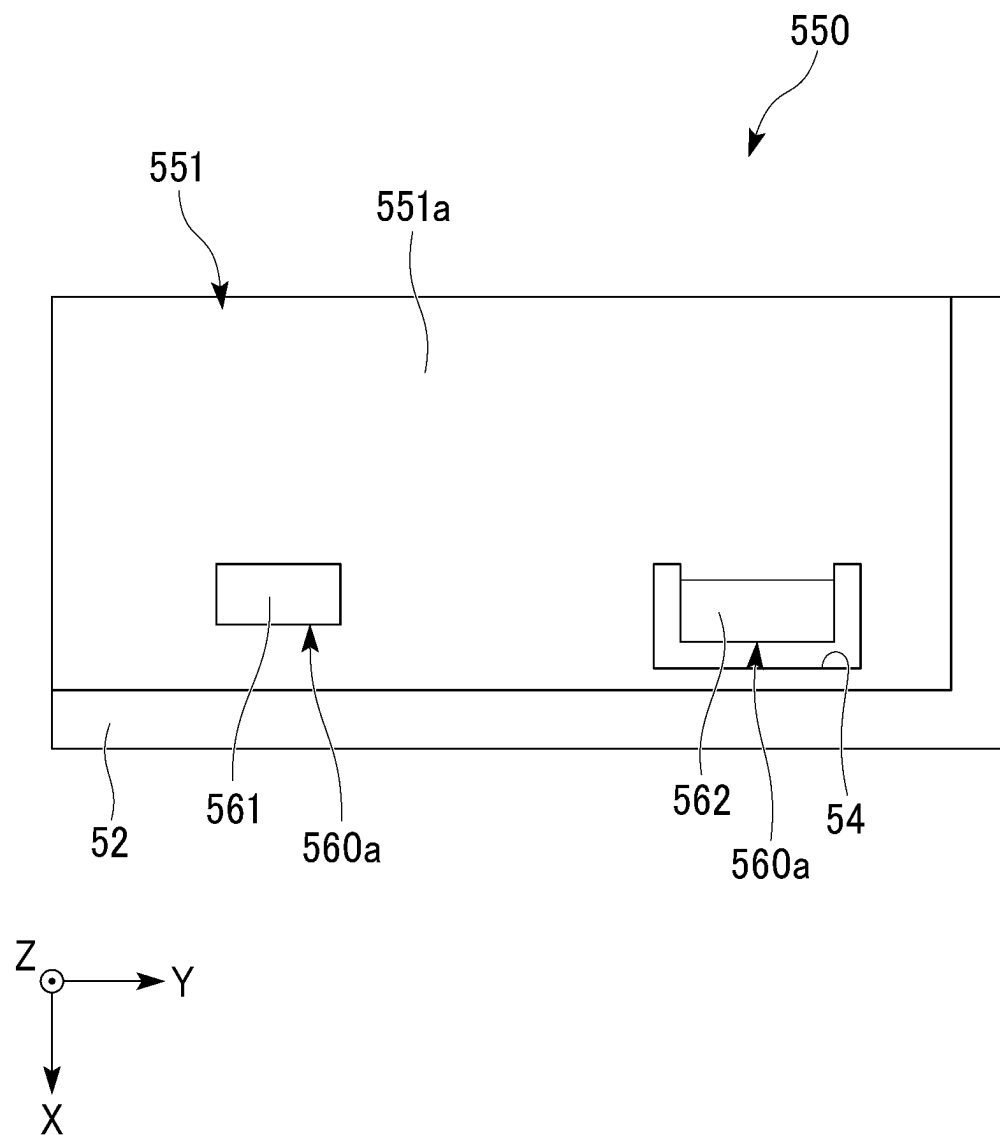


FIG. 15



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HEAT EXCHANGE UNIT**CROSS REFERENCE TO RELATED APPLICATION**

This application is a U.S. national stage application of International Application No. PCT/JP2021/028552, filed on Aug. 2, 2021, the contents of which are incorporated herein by reference.

TECHNICAL FIELD

The present disclosure relates to a heat exchange unit.

BACKGROUND

For example, a grip that is attached to a cabinet of an outdoor unit of an air conditioner that is used during transport or the like is described in Patent Document 1. A method of attaching the grip in Patent Document 1 is done by pressing and fixing grips to side plates of the cabinet by providing open recesses, into which the grips are fit, on a front surface plate and a rear surface plate of the cabinet.

PATENT DOCUMENT

Patent Document 1: Japanese Unexamined Patent Application, First Publication No. H04-359722

In a case similar to the case of the method of attaching a grip as mentioned above, there is a need to sandwich the grip between differing plate members of the cabinet, and the work efficiency of attaching the grip is bad. On the other hand, if openings are provided in the cabinet, and a method of inserting the grip into the openings is adopted, the work efficiency of attaching the grip improves. In such a case however, a problem of the attachment strength of the grips easily declining exists.

SUMMARY

The present disclosure has been made in order to address the problem above, and an object is to provide a heat exchange unit that has a construction where it is possible to ensure an attachment strength of gripping members, while improving a work efficiency of attaching the gripping members.

An embodiment of a heat exchange unit according to the present disclosure is a heat exchange unit of a refrigeration cycle device that includes a housing having openings, and gripping members attached to the openings. Each gripping member of the gripping members has a main body having an inner space that opens to an outside of the housing, and a plurality of claws that protrude from the main body, and that hook to a circumferential edge of the openings from an inside of the housing. The plurality of claws include a plurality of first claw portions, and at least one second claw portion that is provided on an opposite side to the plurality of first claw portions in a predetermined direction that intersects the opening direction of an opening. At least one claw portion out of the plurality of first claw portions and the at least one second claw portion is an elastic claw that is cantilevered by the main body, and the elastic claw is elastically deformable in the predetermined direction with respect to the main body. The plurality of first claw portions include at least one fixed claw, and at least one elastic claw, and the protrusion height of the at least one fixed claw with

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respect to the main body is less than the protrusion height of the at least one elastic claw with respect to the main body.

According to the present disclosure, it is possible to ensure an attachment strength of the gripping members, while improving a work efficiency of attaching the gripping members.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 A schematic block diagram that schematically shows a refrigeration cycle device that is included in an outdoor unit in a first embodiment.

FIG. 2 A perspective view that shows the outdoor unit in the first embodiment.

FIG. 3 A perspective view that shows a part out of parts of the outdoor unit in the first embodiment where a gripping member is attached.

FIG. 4 A perspective view that shows the gripping member in the first embodiment.

FIG. 5 A view of the gripping member in the first embodiment as seen from the front side.

FIG. 6 A cross-sectional view that shows the gripping member in the first embodiment, and corresponds to the cross-sectional line VI-VI in FIG. 5.

FIG. 7 A cross-sectional view that shows the gripping member in the first embodiment, and corresponds to the cross-sectional line VII-VII in FIG. 5.

FIG. 8 A view of the gripping members in the first embodiment as seen from the top side.

FIG. 9 A cross-sectional view that shows a portion of attachment procedures of the gripping member to a housing in the first embodiment.

FIG. 10 A cross-sectional view that shows another portion of the installation procedures of the gripping member to the housing in the first embodiment.

FIG. 11 A perspective view that shows the gripping members in a second embodiment.

FIG. 12 A view of the gripping members in the second embodiment as seen from the top side.

FIG. 13 A view of the gripping members in a third embodiment as seen from the top side.

FIG. 14 A view of the gripping members in a fourth embodiment as seen from the top side.

FIG. 15 A view of the gripping members in a fifth embodiment as seen from the top side.

DETAILED DESCRIPTION

Hereinafter, an embodiment is explained with reference to drawings. The scope of the present disclosure is not limited to the embodiments below, and may be changed appropriately so long as the changes are within the scope of the technical idea of the present disclosure. In order to facilitate better understanding of the various configurations, there are cases where scales and numbers or the like of various configurations in the drawings below differ from scales and numbers of actual configurations.

An X axis, a Y axis, and a Z axis are shown in the drawings as deemed appropriately. The X axis shows a front-back direction of an outdoor unit 10 in the embodiments below. The Y axis shows a width direction of the outdoor unit that is orthogonal to the front-back direction. The Z axis shows a vertical direction. The front-back direction, the width direction, and the vertical direction are directions that mutually cross one another. Hereinafter, the front-back direction is referred to as a “front-back direction X”, the width direction is referred to as a “width direction

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Y”, and the vertical direction is referred to as a “vertical direction Z”. The vertical direction, or a Z axis direction corresponds to a “predetermined direction”.

Out of sides of the front-back direction X, a side to which an arrow of the X axis faces (+X side) is a front side of the outdoor unit. Out of the sides of the front-back direction X, an opposite side to which the arrow of the X axis faces (−X side) is a back side of the outdoor unit. The width direction Y is a left-right direction of the outdoor unit. Out of the sides of the width direction Y, a side to which the arrow of the Y axis faces (+Y side) is a left side of the outdoor unit. Out of the sides of the width direction Y, an opposite side to which the arrow of the Y axis faces (−Y side) is a right side of the outdoor unit. In the description below, the side out of the sides of the width direction Y to which the arrow of the Y axis faces (+Y side), in other words the left side, is referred to as a “one side of the width direction”. The opposite side out of the sides of the width direction Y to which the arrow of the Y axis faces (−Y side) is referred to as an “other side of the width direction”. A side out of the sides of the vertical direction Z to which the arrow of the Z axis faces (+Z side) is a top side. An opposite side out of the sides of the vertical direction Z to which the arrow of the Z axis faces (−Z side) is a bottom side.

First Embodiment

FIG. 1 is a schematic block diagram that schematically shows a refrigeration cycle device 100 which is included in the outdoor unit 10 in a first embodiment. The refrigeration cycle device 100 is a device which utilizes a refrigeration cycle that circulates a refrigerant 40. The refrigeration cycle device 100 in the first embodiment is an air conditioning device. As shown in FIG. 1, the refrigeration cycle device 100 includes the outdoor unit 10, an indoor unit 20, and a circulation path 30. The outdoor unit 10 is disposed outdoors. The indoor unit 20 is disposed indoors. The outdoor unit 10 and the indoor unit 20 are mutually connected by the circulation path 30 that circulates the refrigerant 40. The outdoor unit 10 and the indoor unit 20 make up a heat exchange unit that conducts heat exchange between the units thereof and the surrounding atmosphere.

By the refrigeration cycle device 100 conducting heat exchange between the refrigerant 40 circulating inside the circulation path 30 and the indoor atmosphere where the indoor unit 20 is disposed, the temperature of the indoor atmosphere is adjustable. A refrigerant such as a fluorine based refrigerant with a low global warming potential (GWP: Global Warming Potential), or a hydrocarbon based refrigerant or the like may be mentioned as examples of the refrigerant 40.

The outdoor unit 10 has a housing 11, a compressor 12, a heat exchange device 13, a flow adjustment valve 14, a blower 15, a four-way valve 16, and a controller 17. The compressor 12, the heat exchange device 13, the flow adjustment valve 14, the blower 15, the four-way valve 16, and the controller 17 are housed on an inside of housing 11. The controller 17 controls the various parts of the outdoor unit 10. The controller 17 for example, may be a system controller that regulates the control of the entirety of the refrigeration cycle device 100.

Out of components on the circulation path 30, the compressor 12, the heat exchange device 13, the flow adjustment valve 14, and the four-way valve 16 are provided on a part located on the inside of housing 11. Out of the components on the circulation path 30, the compressor 12, the heat

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exchange device 13, the flow adjustment valve 14, and the four-way valve 16 are connected by the part located on the inside of housing 11.

The four-way valve 16 is provided on a portion connected to a discharge side of the compressor 12 out of components on the circulation path 30. The four-way valve 16 is able to reverse the flow of the refrigerant 40 inside the circulation path 30 by switching over a portion of the path of the circulation path 30. When the path connected by the four-way valve 16 is the path of the four-way valve 16 that is shown by solid lines in FIG. 1, the refrigerant 40 flows in the direction of the circulation path 30 shown by the solid line arrow in FIG. 1. On the other hand, when the path connected by the four-way valve 16 is the path of the four-way valve 16 that is shown by dashed lines in FIG. 1, the refrigerant 40 flows in the direction of the circulation path 30 shown by the dashed line arrow in FIG. 1.

The indoor unit 20 has a housing 21, a heat exchange device 22, and a blower 23. The heat exchange device 22 and the blower 23 are housed on an inside of housing 21. The indoor unit 20 is able to cool the indoor atmosphere the indoor unit 20 is disposed in, through a cooling operation. The indoor unit 20 is able to warm the indoor atmosphere the indoor unit 20 is disposed in, through a heating operation.

When the indoor unit 20 is in the cooling operation, the refrigerant 40 that flows within the circulation path 30 flows in the direction of the solid line arrow shown in FIG. 1. In other words, when the indoor unit 20 is in cooling operation, the refrigerant 40 that flows within the circulation path 30 circulates so as to return to the compressor 12 after passing through the compressor 12, the heat exchange device 13 of the outdoor unit 10, the flow adjustment valve 14, and the heat exchange device 22 of the indoor unit 20 in such an order. During the cooling operation, the heat exchange device 13 inside the outdoor unit 10 functions as a condenser, and the heat exchange device 22 of the indoor unit 20 functions as an evaporator.

On the other hand, when the indoor unit 20 is in the heating operation, the refrigerant 40 that flows within the circulation path 30 flows in the direction of the dashed line arrow shown in FIG. 1. In other words, when the indoor unit 20 is in heating operation, the refrigerant 40 that flows within the circulation path 30 circulates so as to return to the compressor 12 after passing through the compressor 12, the heat exchange device 22 of the indoor unit 20, the flow adjustment valve 14, and the heat exchange device 13 of the outdoor unit 10 in such an order. During the heating operation, the heat exchange device 13 inside the outdoor unit 10 functions as an evaporator, and the heat exchange device 22 of the indoor unit 20 functions as a condenser.

Next, the outdoor unit 10 in the first embodiment is further explained. FIG. 2 is a perspective view that shows the outdoor unit 10. As shown in FIG. 2, the housing 11 is roughly a rectangular box-shape having surfaces that face each of the front-back direction X, the width direction Y, and the vertical direction Z. In the first embodiment, the housing 11 has two blowers 15 that are provided so as to be aligned in the vertical direction Z. The housing 11 has a front surface panel 11a that is attached so as to be detachable. The front surface panel 11a is a sheet metal member. The front surface panel 11a has a front wall portion 11b that configures a portion of a front side (+X side) wall portion in the housing 11, and a side wall portion 11c that configures a portion of the one side of the width direction (+Y side) wall portion in the housing 11. The side wall portion 11c bends to the back side (−X side), from the one side of the width direction of the front wall portion 11b.

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The outdoor unit 10 includes gripping members 50. The gripping members 50 are members to which an operator or the like may hook their hands or fingers to when transporting the outdoor unit 10. The gripping members 50 in the first embodiment are resin members that are integrally formed by a metal mold of injection molding or the like. The resin that the gripping members 50 are configured from may for example be polypropylene or the like. A plurality of the gripping members 50 are provided. The gripping members 50 in the first embodiment are provided on each of four corner portions out of portions of the housing 11 to which the front-back direction X wall portion and the width direction Y wall portion are connected to. The gripping members 50 in the first embodiment are located on the bottom side in the vertical direction Z more than a center of the outdoor unit 10.

FIG. 3 is a perspective view that shows a part out of parts of the outdoor unit 10 where a gripping member 50 is attached. FIG. 4 is a perspective view that shows the gripping member 50. FIG. 5 is a view of the gripping member 50 as seen from the front side (+X side). FIG. 6 is a cross-sectional view that shows the gripping member 50, and corresponds to the cross-sectional line VI-VI in FIG. 5. FIG. 7 is a cross-sectional view that shows the gripping member 50, and corresponds to the cross-sectional line VII-VII in FIG. 5. FIG. 8 is a view of the gripping member 50 as seen from the top side.

The gripping members 50 shown in FIG. 3 to FIG. 8 are gripping members 50 that are provided in the corner portions out of portions of the housing 11 to which the front side (+X side) wall portion and the one side of the width direction (+Y side) wall portion are connected to. The corner portions, to which the gripping members 50 are attached, shown in FIG. 3 to FIG. 8 are provided on the front surface panel 11a. Although the four gripping member 50 are each attached in differing locations and orientations, each have the same construction. For such reason, the relative positional relationship and the like of the gripping member 50 attached to the front surface panel 11a, as a representative of the four gripping members 50, is described in the explanation of the gripping member 50 below.

As shown in FIG. 3, the gripping member 50 is attached to an opening 11d that is provided in the housing 11. The opening 11d penetrates the wall portion that configures the housing 11. The opening 11d shown in FIG. 3 is provided so as to extend over the front wall portion 11b and the side wall portion 11c in the front surface panel 11a. An opening direction of the opening 11d in the first embodiment is a direction that is orthogonal to the vertical direction Z. In other words, the vertical direction Z is orthogonal to the opening direction of the opening lid. The opening lid shown in FIG. 3 opens to the front side (+X side) and to the one side of the width direction (+Y side). The opening 11d is covered by attaching the gripping member 50. A protrusion 11e that protrudes to the inside of housing 11 is provided on an inner edge of the opening 11d, as shown in FIG. 6 and FIG. 7. The protrusion 11e is at least provided on the portions out of portions of the inner edge of the opening 11d that face a plurality of claws 60, to be mentioned later on.

As shown in FIG. 4 to FIG. 8, the gripping member 50 has a main body 51 and a frame body 52. The main body 51 is roughly a rectangular box-shape that opens to the front side (+X side) and to the one side of the width direction (+Y side). As shown in FIG. 6 and FIG. 7, the main body 51 is housed on the inside of housing 11.

The main body 51 has an upper wall portion 51a, a bottom wall portion 51b, an inner wall portion 51c, and a side wall

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portion 51d. Out of wall portions of the main body 51, the upper wall portion 51a is the wall portion that is located on the top side. Out of wall portions of the main body 51, the bottom wall portion 51b is the wall portion that is located on the bottom side. Out of wall portions of the main body 51, the inner wall portion 51c is the wall portion located on the back side (-X side). Out of wall portions of the main body 51, the side wall portion 51d is the wall portion located on the other side of the width direction (-Y side).

As shown in FIG. 4, the upper wall portion 51a has through holes 54 that penetrate the upper wall portion 51a in the vertical direction Z. A pair of through holes 54 are provided at an interval in the width direction Y in the first embodiment. The through holes 54 are provided on the front side (+X side) portion of the upper wall portion 51a. As shown in FIG. 8, the through holes 54 are roughly long rectangular shapes in the width direction Y, as seen from the vertical direction Z. Edges of the front side of the through holes 54 are disposed so as to be spaced apart in the back side (-X side) more than the edge of the front side of the upper wall portion 51a.

As shown in FIG. 6 and FIG. 7, the frame body 52 is connected to the opening edge 51e of the main body 51. The frame body 52 is a frame shape along the opening edge 51e of the main body 51. The frame body 52 protrudes from the opening edge 51e of the main body 51 to the outside over the entire circumference of the opening edge 51e. The frame body 52 is located on the outside of housing 11. Out of parts of the frame body 52, the parts that protrude to the outside more than the main body 51 are coming into contact with, and hook onto, a circumferential edge of the opening 11d from an outside of housing 11, out of the outside surfaces of the housing 11. The circumferential edge of the opening 11d is sandwiched between the plurality of claws 60 to be mentioned later on, and the frame body 52.

As shown in FIG. 4, the frame body 52 has a first portion 52a that is connected to front side (+X side) of the main body 51, and a second portion 52b that is connected to the one side of the width direction (+Y side) of the main body 51. The first portion 52a protrudes to both sides of the vertical direction Z and to the other side of the width direction (-Y side) more than the main body 51. The second portion 52b extends to the back side (-X side) from an end portion of the one side of the width direction of the first portion 52a. The second portion 52b protrudes to both sides of the vertical direction Z and the back side more than the main body 51.

The inside of frame body 52 is an insertion opening 52c that is connected to the inside of main body 51. The frame body 52 has a cover portion 52d that covers atop side portion out of portions of the opening of the main body 51. A hand insertion space 53 (inside space) is configured by the main body 51 and the frame body 52. The hand insertion space 53 is a space into which the operator is able to hook their hands or fingers to when transporting the outdoor unit 10. The hand insertion space 53 opens to the outside of housing 11 via the insertion opening 52c. The hand insertion space 53 is a space that is provided on the inside of the main body 51. The operator transporting the outdoor unit 10 inserts their hands or fingers in the hand insertion space 53 and holds the cover portion 52d. Therefore, it is possible for the operator to lift the outdoor unit 10 by holding onto the gripping member 50.

As shown in FIG. 6, the gripping member 50 has the plurality of claws 60. The plurality of claws 60 protrude from the main body 51. The plurality of claws 60 are located on the inside of housing 11. The plurality of claws 60 are disposed so as to face the circumferential edge of the

openings 11d in the inside of housing 11. The plurality of claws 60 are disposed so as to face an inner end portion of the protrusion 11e that is provided in the opening 11d. The plurality of claws 60 hook onto the circumferential edge of the openings 11d from the inside of the housing 11.

As shown in FIG. 5, the plurality of the claws 60 include first claw portions 60a, and second claw portions 60b. The first claw portions 60a are claws 60 that are provided on the top side of the main body 51. The second claw portions 60b are claws 60 that are provided on the bottom side of the main body 51. In other words, the second claw portions 60b are provided on the opposite side of the first claw portions 60a in the vertical direction Z. Each of a plurality of first claw portions 60a and of second claw portions 60b are provided in the first embodiment. The plurality of first claw portions 60a protrude to the top side from the upper wall portion 51a of the main body 51. The plurality of the second claw portions 60b protrude to the bottom side from the bottom wall portion 51b of the main body 51.

The first claw portions 60a include a fixed claw 61, and elastic claws 62. The fixed claw 61 and the elastic claws 62 protrude to the top side from the upper wall portion 51a of the main body 51. As shown in FIG. 8, the fixed claw 61 in the first embodiment is provided between a pair of through holes 54 on a portion out of portions of the upper wall portion 51a in the width direction Y. The fixed claw 61 is located on a center portion of the upper wall portion 51a in the width direction Y. A bottom end portion of the fixed claw 61 is entirely connected to the upper wall portion 51a. As shown in FIG. 4 and FIG. 6, the fixed claw 61 is roughly a triangular shape, as seen from the width direction Y. The protrusion height of the fixed claw 61 with respect to the main body 51 becomes larger as the fixed claw 61 approaches the front side (+X side) of the main body 51. The protrusion height of the fixed claw 61 with respect to the main body 51 is a vertical direction Z dimension of the fixed claw 61, and represents a distance between a top surface of the upper wall portion 51a and a top end portion of the fixed claw 61 in the vertical direction Z. The top end portion of the fixed claw 61 in the first embodiment is located more to the bottom side than the top end portion of the frame body 52.

As shown in FIG. 6, the fixed claw 61 has a first facing surface 61f that faces the circumferential edge of the openings 11d, and a first inclined surface 61i. The first facing surface 61f faces the front side (+X side). The first facing surface 61f is a surface that is roughly parallel with the vertical direction Z. For example, although not shown in the figures, the first facing surface 61f is a surface that slightly inclines towards a direction located in the back side with respect to the vertical direction Z as the first facing surface 61f approaches the top side. The first facing surface 61f faces the inner end portion of protrusion 11e in the first embodiment. The first facing surface 61f contacts the inner end portion of protrusion 11e. The first inclined surface 61i is a surface that faces the back side, and faces an inclined top side. A top end portion of the first inclined surface 61i is connected to a top end portion of the first facing surface 61f. The first inclined surface 61i becomes located on the top side as the first inclined surface 61i moves towards the front side.

When the first facing surface 61f is hooked onto the protrusion 11e, the fixed claw 61 is hooked from the inside of housing 11 to the circumferential edge of the openings 11d. A portion out of portions of the circumferential edge of the openings 11d that is located on the top side is sandwiched in the front-back direction X by a portion that protrudes to the top side more than the main body 51 out of portions of the frame body 52, and by the fixed claw 61.

As shown in FIG. 4, the elastic claws 62 in the first embodiment are provided in a pair that sandwiches the fixed claw 61 in the width direction Y on the upper wall portion 51a. Each elastic claw 62 in the pair of elastic claws 62 has the same shape. The pair of elastic claws 62 are each disposed inside the pair of through holes 54. The pair of elastic claws 62 are each cantilevered by an inner edge of each of the pair of through holes 54. An end of each of the elastic claw 62 in the back side (-X side) is connected to a portion out of portions of the through holes 54 that is located to the back side of the inner edge of each of the through holes 54. In other words, the elastic claws 62 in the first embodiment are connected from the opening 11d out of the inner edges of through hole 54, to an edge of a far side of through hole 54. Therefore, each of the elastic claws 62 is cantilevered by the main body 51. A gap is provided on the edges of both sides of the width direction between the elastic claws 62 and the through holes 54, and on the edge of the front side between the elastic claws 62 and the through holes 54.

As shown in FIG. 7, an elastic claw 62 of the elastic claws 62 has a support portion 62a and a claw main body 62b. The support portions 62a in the first embodiment are plate shapes that have plate surfaces which face the vertical direction Z. The support portions 62a are located inside the through holes 54. The support portions 62a extend from the portion out of portions of inner edges of each of the through holes 54 that is located on the back side (-X side) to the front side (+X side). An end of the support portions 62a front side is disposed so as to face the back side of the portion that is located on the front side of the through holes 54 out of portions of the inner edges of the through holes 54, with a gap in between.

The claw main bodies 62b protrude to the top side from the portion in the front side (+X side) of the support portions 62a. The claw main bodies 62b protrude to the top side more than the through holes 54. The claw main bodies 62b are roughly triangular shapes, as seen from the width direction Y. Protrusion heights of the claw main bodies 62b with respect to the main body 51 become larger as the claw main bodies 62b approach the front side of the main body 51. The protrusion heights of the claw main bodies 62b with respect to the main body 51 are vertical direction Z dimensions of the claw main bodies 62b, and are protrusion heights of the elastic claws 62 with respect to the main body 51. The protrusion heights of the elastic claws 62 with respect to the main body 51 represent distances between the top surface of the upper wall portion 51a and top end portions of the elastic claws 62 in the vertical direction Z.

As shown in FIG. 5, top end portions of the elastic claws 62 in the first embodiment, in other words the top end portions of the claw main bodies 62b, are located on the top side more than the top end portion of the fixed claw 61. In other words, the protrusion height of the fixed claw 61 with respect to the main body 51 is less than the protrusion heights of the elastic claws 62 with respect to the main body 51. The top end portions of the elastic claws 62 are located in the bottom side more than the top end portion of the frame body 52.

As shown in FIG. 7, the elastic claws 62 have second facing surfaces 62f that face the circumferential edge of the openings 11d, and second inclined surfaces 62i. The second facing surfaces 62f face the front side (+X side). The second facing surfaces 62f are surfaces that are roughly parallel with the vertical direction Z. For example, although not shown in the figures, the second facing surfaces 62f are surfaces that slightly incline towards a direction located in the back side

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with respect to the vertical direction Z as the second facing surfaces **62f** approach the top side. The second facing surfaces **62f** are connected to the top side of the support portions **62a** front side surface. The second facing surfaces **62f** in the first embodiment face inner end portion of the protrusion **11e**. The second facing surfaces **62f** are in contact with the inner end portion of the protrusion **11e**. The second inclined surfaces **62i** are surfaces that face the back side (−X side), and incline towards the top side. The second inclined surfaces **62i** top sides are connected to the second facing surfaces **62f** top sides. The second inclined surfaces **62i** become located on the top side as the second inclined surfaces **62i** move towards the front side.

When the second facing surfaces **62f** are hooked to the protrusion **11e**, the elastic claws **62** are hooked from the inside of the housing **11** to the circumferential edge of the openings **11d**. A portion out of portions of the circumferential edge of the openings **11d** that is located on the top side is sandwiched in the front-back direction X by a portion that protrudes to the top side more than the main body **51** out of portions of the frame body **52**, and by the elastic claws **62**.

As shown in FIG. 8, a dimension of the elastic claws **62** in the width direction Y is greater than a dimension of the fixed claw **61** in the width direction Y. Surfaces of the elastic claws **62** front sides (+X side) are located slightly more to the front side than a surface of the fixed claw **61** front side. In other words, the second facing surfaces **62f** are located on a side closer to the circumferential edge of the openings **11d** than the first facing surface **61f**.

The elastic claws **62** deform elastically in the vertical direction Z with respect to the main body **51**. Specifically, by making connection portions of the inner edges of the through holes **54** in the support portions **62a** elastically deformable in the vertical direction Z, it is possible to have the elastic claws **62** elastically deform in the vertical direction Z with the connection portions of the support portions **62a** and the inner edges of the through holes **54** as a pivot.

As shown in FIG. 5, the second claw portions **60b** include fixed claws **63**. The fixed claws **63** protrude to the bottom side from the bottom wall portion **51b** of the main body **51**. A pair of fixed claws **63** in the first embodiment is provided in the width direction Y with a space in between. The pair of fixed claws **63** are each disposed in a location that overlaps with the pair of elastic claws **62** as seen from the vertical direction Z. Top end portions of the fixed claws **63** are entirely connected to the bottom wall portion **51b**. As shown in FIG. 7, the fixed claws **63** are roughly shaped as isosceles triangles, as seen from the width direction Y. Protrusion heights of the fixed claws **63** with respect to the main body **51**, in other words the protrusion heights of the second claw portions **60b** with respect to the main body **51**, are greater than the protrusion heights of the elastic claws **62** with respect to the main body **51**. The protrusion heights of the fixed claws **63** with respect to the main body **51** in the first embodiment are greater than the protrusion heights of elastic claws **62** with respect to the main body **51**. The protrusion heights of the fixed claws **63** with respect to the main body **51** are vertical direction Z dimensions of the fixed claws **63**, and represent distances between a bottom surface of the bottom wall portion **51b** and a bottom end portions of fixed claws **63** in the vertical direction Z. The bottom end portions of fixed claws **63** in the first embodiment are located more to the bottom side than a bottom end portion of frame body **52**.

The fixed claws **63** have third facing surfaces **63f** that face the circumferential edge of the openings **11d**. The third facing surfaces **63f** face the front side (+X side). The third

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facing surfaces **63f** are surfaces that incline towards a direction located in the back side (−X side) with respect to the vertical direction Z as the third facing surfaces **63f** approach the bottom side. The third facing surfaces **63f** face the inner end portion of the protrusion **11e** in the first embodiment. The third facing surfaces **63f** contact the inner end portion of the protrusion **11e**. A surface on the back side of the fixed claws **63** is symmetrically disposed to the third facing surfaces **63f** in the front back direction X.

When the third facing surfaces **63f** are hooked onto the protrusion **11e**, the third facing surfaces **63f** are hooked from the inside of the housing **11** to the circumferential edge of the openings **11d**. A portion out of portions of the circumferential edge of the openings **11d** that is located on the bottom side is sandwiched in the front-back direction X by a portion that protrudes to the bottom side more than the main body **51** out of portions of the frame body **52**, and by the fixed claws **63**.

FIG. 9 is a cross-sectional view that shows a portion of attachment procedures of the gripping member **50** to the housing **11**. FIG. 10 is a cross-sectional view that shows another portion of the attachment procedures of the gripping member **50** to the housing **11** in the first embodiment. The operator attaching the gripping members **50** to the housing **11** brings the gripping members **50** close to the openings **11d** from an outside of the housing **11**. The operator then tilts the gripping member **50** with respect to the vertical direction Z, and inserts the fixed claws **63** provided on a bottom side portion of the main body **51** and on the bottom side of the main body **51** to an inside of the openings **11d**.

In a state where the fixed claws **63** are hooked to circumferential edge of the openings **11d** on the inside of the housing **11**, the operator inserts the top side portion of the main body **51** as shown in FIG. 9 to the inside of the openings **11d**, straightening out the tilt of the gripping members **50** with respect to the vertical Z direction. In FIG. 9 and in FIG. 10, although a state where the top side portion of the main body **51** is moved straight in the back side (−X side) and is inserted to the inside of the opening **11d**, it is possible to insert the top side portion of the main body **51** to the inside of the opening **11d** by rotating the gripping members **50**, with the bottom side portion of the gripping members **50** that are hooked onto the opening **11d** as a pivot.

After inserting the top side portion of the main body **51** to the inside of the opening **11d**, the second inclined surfaces **62i** of the elastic claws **62** contact the circumferential edge of the top side of the opening **11d**. In this state, when the gripping members **50** are further pressed into the openings **11d**, as shown in FIG. 10, the second inclined surfaces **62i** are pressed by the circumferential edge of the openings **11d**, and the elastic claws **62** deform in the direction (−Z direction) in which the elastic claws **62** are push-inserted to the inside of the main body **51**, where the connection portions of the support portions **62a** and the inner edges of the through holes **54** act as pivots. Therefore, it is possible to decrease the protrusion heights of elastic claws **62** with respect to the main body **51**, and to easily insert the elastic claws **62** to the inside of the openings **11d**.

When the main body **51** is further pressed to the inside of opening **11d**, the elastic claws **62** entirely pass through the opening **11d**, causing the elastically deformed support portions **62a** to restore, and the elastic claws **62** return to the original state of the elastic claws **62**. Therefore, the second facing surfaces **62f** of the elastic claws **62** are in a state where the second facing surfaces **62f** of the elastic claws **62** face the inner end portion of the protrusion **11e**, and the

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elastic claws **62** are hooked onto the circumferential edge of the openings **11d** from the inside of the housing **11**.

As previously mentioned, when the gripping members **50** are push-inserted to the inside of the openings **11d**, as with elastic claws **62**, the fixed claw **61** provided on the top side of the main body **51** also contacts the edge of the top side of the openings **11d**. In such case, the protrusion height of the fixed claw **61** is less than the protrusion heights of elastic claws **62**. As such, when push-inserting a gripping member **50** of the gripping members **50** to the inside of opening **11d**, by having at least one of the inner edge of the opening **11d** or the fixed claw **61** slightly deform elastically, the fixed claw **61** is inserted into the housing **11** through the opening lid. After the fixed claw **61** passes through the opening lid, the elastically deformed portion is restored, and the first facing surface **61f** of the fixed claw **61** faces the inner end portion of protrusion **11e**. Therefore, the fixed claw **61** is hooked onto the circumferential edge of the openings **11d** from the inside of the housing **11**.

As shown above, it is possible to attach the gripping members **50** to the openings **11d** of the housing **11** by push-inserting the gripping members **50** to the inside of the openings **11d** from the outside of the housing **11**. In a state where the gripping members **50** are attached to the housing **11**, the circumferential edge of the openings **11d** is sandwiched by the plurality of claws **60** and by the frame body **52**. Therefore, it is possible to suppress the gripping members **50** from coming off of the opening **11d**.

In a case where conventional gripping members are inserted and attached to the openings **11d** from the outside of housing **11**, as with the gripping members **50** mentioned above, after inserting the claws provided on one side in the vertical direction **Z** to the inside of the openings **11d**, all of the claws that are provided on the other side in the vertical direction **Z** are pressed into the inside of the openings **11d**, as with fixed claws **61** previously mentioned. In other words, at least one of the inner edges of the openings **11d** and all of the claws provided in the other side in the vertical direction **Z** is slightly deformed elastically when push-inserting the above claws to the inside of openings **11d**. In such case, when the height of protrusions of the elastically deformable claws that are push-inserted are large, it is not possible to push and insert the claws with the claws being hooked onto the inner edges of the openings **11d** to the inside of the openings **11d**, and there is a need to decrease the size of the protrusion heights of the claws. As such, when push-inserting and attaching the conventional gripping members to the inside of openings **11d** from the outside of housing **11**, it is difficult for the claws to have enough protrusion heights, and so attaching the gripping members firmly with respect to the housing **11** is difficult.

In contrast to the above, the elastic claws **62** according to the first embodiment are provided so as to be cantilevered by the main body **51**. The elastic claws **62** are elastically deformable in the vertical direction **Z** with respect to the main body **51**. As such, by elastically deforming the elastic claws **62** as previously mentioned, it is possible to insert the gripping members **50** into the openings **11d** in a state where the protrusion heights of the elastic claws **62** with respect to the main body **51** are smaller. After the elastic claws **62** pass through the openings **11d**, the elastic claws **62** restore after deforming, and the protrusion heights of elastic claws **62** with respect to the main body **51** return to the protrusion heights before deforming. As such, even if the protrusion heights of elastic claws **62** are made to be large to some extent, by push-inserting the gripping members **50** into the openings **11d**, it is possible to easily insert the elastic claws

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62 into the housing **11**. Therefore, it is possible to make each of protrusion heights of claws **60** provided on both sides in the vertical direction **Z** of the main body **51** larger, and it is possible to firmly attach the gripping members **50** to the openings **11d**.

From the above, according to the first embodiment, it is possible to ensure an attachment strength of the gripping members **50**, along with improving work efficiency of the gripping members **50**. When attaching the gripping members **50**, it is possible for the operator to easily confirm the attachment of the gripping members **50** to the openings **11d** by sounds or feeling and so on that result when the elastic claws **62** restore after deforming. As such, it is possible to increase the work efficiency of the gripping members **50**.

By push-inserting the gripping members **50** into the openings **11d** from the outside of the housing **11**, by providing the openings **11d** in the housing **11** in order to attach the gripping members **50** to the housing **11**, it is possible to attach the gripping members **50** to any location of the housing **11**. As such, compared to a case where the gripping members **50** are held by being sandwiched between two or more members that configure the housing **11**, it is possible to improve a degree of freedom of design of attachment locations of the gripping members **50** in the housing **11**. Therefore, it is possible to attach the gripping members **50** to locations where it is easier to hold the gripping members **50**, out of locations of the housing **11**.

The elastic claws **62** according to the first embodiment are the first claw portions **60a**. The protrusion heights of the second claw portions **60b** with respect to the main body **51** are greater than the protrusion heights of the elastic claws **62** with respect to the main body **51**. As such, it is possible to increase the protrusion heights of the second claw portions **60b** to be greater than or equal to the protrusion heights of the elastic claws **62**, protrusion heights of that can be relatively great, as previously mentioned. As such, it is possible to appropriately increase each of the protrusion heights of the elastic claws **62**, which are the first claw portions **60a**, and of the second claw portions **60b**. Therefore, it is possible to firmly attach the gripping members **50** to the openings **11d**.

According to the first embodiment, the main body **51** has through holes **54** that are configured to penetrate the wall portion of the main body **51**. The elastic claws **62** are cantilevered by the inner edges of the through holes **54**. As such, it is possible to easily have the elastic claws **62** be cantilevered by the main body **51**. It is also possible to easily have the elastic claws **62** elastically deform via the through holes **54**.

According to the first embodiment, the elastic claws **62** in the first embodiment are connected to edges, out of the inner edges of through hole **54**, of the sides that are far from the opening **11d**. As such, compared to a case where the elastic claws **62** are connected to the edges closer to the opening **11d** out of the inner edges of through hole **54**, it is easier to have the elastic claws **62** be in a separable orientation from the inner edges of openings **11d** when push-inserting the gripping members **50** into the openings lid. Therefore, it is possible to elastically deform the elastic claws **62**, and it is possible to easily attach the gripping members **50** to the openings **11d**.

According to the first embodiment, the elastic claws **62** are provided on the top side of the main body **51** in the vertical direction. As such, the through holes **54** are also provided on the upper wall portion **51a** of the main body **51**. Therefore, compared to a case where the through holes **54** are provided on the bottom wall portion **51b** of the main

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body 51, it is possible to suppress foreign bodies such as rain water or the like from entering through the through holes 54 to the housing 11.

According to the first embodiment, the plurality of first claw portions 60a are provided. The plurality of first claw portions 60a include the fixed claws 61 and the elastic claws 62. As such, along with the elastic claws 62, it is possible to also hook the fixed claws 61 to the circumferential edge of the opening 11d. As such, it is possible to attach the gripping members 50 to the openings 11d more firmly. The protrusion heights of the fixed claws 61 with respect to the main body 51 are less than the protrusion heights of the elastic claws 62 with respect to the main body 51. Therefore, even if the fixed claws 61 are not configured so as to be cantilevered, such as the case of the elastic claws 62, it is possible to push-insert the fixed claws 61 into the opening 11d. By providing the fixed claws 61 so as not to be cantilevered, a force applied to the gripping members 50 when the outdoor unit 10 is transported or the like is not only absorbed by the elastic claws 62, but by the fixed claws 61 as well. As such, it is possible to prevent excess load from being applied to the elastic claws 62. Therefore, it is possible to prevent concentration of stresses on portions out of the elastic claws 62 that are connected to the through holes 54, in other words a base portion of the support portions 62a, and it is possible to prevent the elastic claws 62 from being damaged.

According to the first embodiment, the second facing surfaces 62f of the elastic claws 62 are located on a side closer to the circumferential edge of the openings 11d than the first facing surfaces 61f of the fixed claws 61. As such, in a case where the gripping members 50 are pulled to the outside of the housing 11 during transport or the like of the outdoor unit 10, it is possible to have the second facing surfaces 62f of the elastic claws 62 to come into contact with the circumferential edge of openings 11d before the first facing surface 61f of the fixed claws 61. As such, it is possible to prevent the fixed claws 61 from coming out from the openings 11d. Therefore, it is possible to attach the gripping members 50 more securely to the openings 11d than the conventional gripping members. When attaching the gripping members 50 to the openings 11d, the fixed claws 61 pass through the openings 11d before the elastic claws 62 pass through the openings 11d. As such, after passing the elastic claws 62 through the openings 11d and confirming that the elastic claws 62 have restored after deforming through sounds or feeling or the like, the fixed claws 61 are in a state where the fixed claws 61 have passed through the openings 11d. Therefore, by being able to confirm that the elastic claws 62 have passed through the openings 11d through sounds or feeling or the like, it is easy to confirm appropriate attachment of the gripping members 50 with respect to the housing 11.

According to the first embodiment, the plurality of first claw portions 60a include a plurality of elastic claws 62. As such, it is possible to firmly attach the gripping members 50 to the openings 11d by the plurality of elastic claws 62. Compared to a case where a dimension of one of the elastic claws 62 in the width direction Y is made larger so as to improve the attachment strength of the gripping members 50, it is possible to make the dimension of each of the elastic claws 62 in the width direction Y smaller, making it easy to elastically deform each of the elastic claws 62. Therefore, it is possible to easily attach the gripping members 50 to the openings lid.

According to the first embodiment, the gripping members 50 have the frame body 52 along the frame of the opening edge 51e of the main body 51. The frame body 52 is located

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on the outside of the housing 11, and sandwiches the circumferential edge of the openings 11d in between with the plurality of claws 60. As such, it is possible to stably sandwich the circumferential edge of the openings 11d, and to more firmly attach the gripping members 50 to the openings 11d by the frame body 52 and the plurality of claws 60.

According to the first embodiment, the heat exchange unit that includes the gripping members 50 provided with the elastic claws 62 is the outdoor unit 10. The mass of the outdoor unit 10 is greater, compared to the indoor unit 20. As such, the outdoor unit 10 is harder to transport, compared to the indoor unit 20. With respect to the above, by providing the gripping members 50 that ensures the previously mentioned attachment strength, it is possible to easily transport the outdoor unit 10 via the gripping members 50.

According to the first embodiment, top end portions of first claw portions 60a are located in the bottom side more than top end portions of frame body 52. Bottom end portions of the second claw portions 60b are located more to the bottom side than the bottom end portions of frame body 52. As such, the first claw portions 60a are hidden and are not visible by the frame body 52, as seen from the front side (+X side) of the gripping members 50. On the other hand, the bottom end portions of the second claw portions 60b protrude to the bottom side more than the frame body 52. As such, the bottom end portions of the second claw portions 60b are visible from the front side before attaching the gripping members 50 to the housing 11. Therefore, the operator attaching the gripping members 50 is able to easily grasp an orientation of the gripping members 50 when attaching the gripping members 50 to the housing 11 since the second claw portions 60b that protrude more than the frame body 52 are visible.

According to the first embodiment, a surface out of surfaces of the elastic claws 62 that is located on an opposite side to the surface where the opening 11d is located on, is the second inclined surfaces 62i which is inclined with respect to the vertical direction Z. As such, when push-inserting the gripping members 50 into the openings lid, the circumferential edge of the openings 11d contact the second inclined surfaces 62i. Therefore, when push-inserting the gripping members 50 into the openings 11d, it is difficult for the elastic claws 62 to get hooked onto the circumferential edge of the openings 11d, and it becomes easier to elastically deform the elastic claws 62. It is also difficult to apply an excessive force to the circumferential edge of the openings 11d. Therefore, it is possible to prevent deformation of the openings 11d.

Second Embodiment

FIG. 11 is a perspective view that shows gripping members 250 in a second embodiment. FIG. 12 is a view of a gripping member 250 in the second embodiment as seen from the top side. In the explanation below, configurations similar to the configurations previously mentioned have the same reference symbols appropriated, and thus the explanations thereof are omitted.

As shown in FIG. 11 and in FIG. 12, the gripping members 250 in the second embodiment have an upper wall portion 251a of main body 251. In contrast to the upper wall portion 51a of the first embodiment, the upper wall portion 251a of the main body 251 only has one through hole 54 in a center portion of the upper wall portion 251a in the width

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direction Y. Another configuration of the main body **251** is similar to another configuration of the main body **51** in the first embodiment.

A plurality of first claw portions **260a** in the gripping members **250** includes two fixed claws **261** and one elastic claw **262**. The one elastic claw **262** is provided in the one through hole **54**. The two fixed claws **261** are disposed so as to sandwich the one elastic claw **262** in the width direction Y. Another configuration of the fixed claws **261** is similar to another configuration of the fixed claws **61** of the first embodiment. Another configuration of an elastic claw **262** is similar to another configuration of the elastic claws **62** of the first embodiment. Another configuration of the gripping members **250** is similar to another configuration of the gripping members **50** of the first embodiment.

According to the second embodiment, the plurality of first claw portions **260a** includes the plurality of fixed claws **261** and the one elastic claw **262**. As such, it is possible to appropriately receive the stresses that are applied to the elastic claw **262** by the plurality of fixed claws **261**.

Therefore, it is possible to prevent damage to the elastic claw **262**, as well as appropriately and safely attach the gripping members **250** to the openings lid. When attaching the gripping members **250**, when it is possible to confirm through sounds or feeling or the like whether the one elastic claw **262** has restored inside of housing **11** after deforming and is hooked onto the circumferential edge of the opening **11d**, it is then possible to confirm that the gripping members **250** are correctly attached to the openings **11d**. In other words, if it is possible to insert one elastic claw **262** into the housing **11** correctly, there is no need to confirm whether the other elastic claws **262** have been inserted into the housing **11** correctly or not. As such, it is possible to prevent attachment of the gripping members **250** to the housing **11** in a state where the elastic claws **262** are elastically deformed on the inside of opening **11d**. Therefore, it is possible to appropriately and easily attach the gripping members **250** to the openings **11d**.

Third Embodiment

FIG. **13** is a view of gripping members **350** in a third embodiment as seen from the top side. In the explanation below, configurations similar to the configurations previously mentioned have the same reference symbols appropriated, and thus the explanations thereof are omitted.

As shown in FIG. **13**, except that two through holes **54** have different dispositions on an upper wall portion **351a**, a main body **351** in the gripping members **350** of the third embodiment has a similar configuration as the configuration of the main body **51** in the first embodiment.

A plurality of first claw portions **360a** includes one fixed claw **361**, and two elastic claws **362**. The one fixed claw **361** is disposed adjacently to the two elastic claws **362** in the other side of the width direction ($-Y$ side). The two elastic claws **362** are disposed adjacently in the width direction Y. A dimension of the elastic claws **362** in the width direction Y is less than a dimension of the fixed claw **361** in the width direction Y. As such, it is possible for the elastic claws **362** to elastically deform when attaching to the gripping members **350**. Therefore, it is possible to easily attach the gripping members **350**.

Another configuration of the fixed claw **361** is similar to another configuration of the fixed claw **61** of the first embodiment. Another configuration of the elastic claws **362** is similar to another configuration of the elastic claws **62** of the first embodiment. Another configuration of the gripping

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members **350** is similar to another configuration of the gripping members **50** of the first embodiment.

Fourth Embodiment

FIG. **14** is a view of gripping members **450** in a fourth embodiment as seen from the top side. In the explanation below, configurations similar to the configurations previously mentioned have the same reference symbols appropriated, and thus the explanations thereof are omitted.

As shown in FIG. **14**, a plurality of first claw portions **460a** includes only two elastic claws **62** in gripping members **450** of the fourth embodiment. Unlike the first embodiment, the plurality of the first claw portions **460a** do not include the fixed claw **61**. As such, compared to a case where the fixed claws **61** are provided, the gripping members **450** are easier to make. Another configuration of the gripping members **450** is similar to another configuration of the gripping members **50** of the first embodiment.

Fifth Embodiment

FIG. **15** is a view of gripping members **550** in a fifth embodiment as seen from the top side. In the explanation below, configurations similar to the configurations previously mentioned have the same reference symbols appropriated, and thus the explanations thereof are omitted.

As shown in FIG. **15**, except that the one through hole **54** is disposed in the one side of the width direction ($+Y$ side) on an upper wall portion **551a**, a main body **551** in the gripping members **550** of the fifth embodiment has a similar configuration as the configuration of the main body **251** in the second embodiment. A plurality of first claw portions **560a** each only include one fixed claw **561** and one elastic claw **562**. As such, by decreasing the number of first claw portions **560a** and making the gripping members **550** easy to make, the attachment strength of the gripping members **550** is improved by the fixed claw **561** and the elastic claw **562**. The one fixed claw **561** is disposed in the other side of the width direction ($-Y$ side) of the one elastic claw **562**.

Another configuration of the fixed claw **561** is similar to another configuration of the fixed claw **61** in the first embodiment. Another configuration of the elastic claw **562** is similar to another configuration of the elastic claws **62** in the first embodiment. Another configuration of the gripping members **550** is similar to another configuration of the gripping members **50** in the first embodiment.

Although embodiments of the present disclosure are described above, the disclosure is not limited to the aforementioned various configurations of the embodiments, and the configurations and methods mentioned below may be adopted.

So long as the claw portions at least include one first claw portion, and at least include one second claw portion, any other claw portions may be included. So long as one claw portion out of the at least one first claw portion included and the at least one second claw portion included is an elastic claw, the other first claw portion or the other second claw portion may be any claw portion. Each of the first claw portion and the second claw portion may include elastic claws. All of the first claw portions and all of the second claw portions may be elastic claws. Shapes of various claw portions may be any shape.

Protrusion heights of the various claw portions with respect to the main body are not limited in any way. A protrusion height of the first claw portion with respect to the main body may be greater than a protrusion height of the

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second claw portion with respect to the main body. A protrusion height of the elastic claw with respect to the main body may be greater than a protrusion height of the second claw portion with respect to the main body. A relationship between protrusion heights of the various claw portions and the frame body are not limited in any way. So long as the elastic claws are cantilevered by the main body, the elastic claws may be connected to the main body in any manner. Predetermined directions in which the first claw portions and the second claw portions are provided on need not be vertical directions, so long as the directions are orthogonal to the opening direction of the openings.

The heat exchange unit of the present disclosure that is provided with gripping members may be an indoor unit of a refrigeration cycle device. In such a case, an indoor unit for example, may be a floor-mounted indoor unit, or a ceiling-mounted indoor unit or the like. The indoor refrigeration cycle device that includes the heat exchange unit of the present disclosure is not limited to an air conditioning device, so long as the device utilizes a refrigeration cycle where a refrigerant is circulated. The refrigeration cycle device may be a heat pump water heater or the like.

The various configurations and methods described in the above description of the present disclosure may be appropriately combined, as long as the configurations and methods thereof do not mutually contradict one another.

The invention claimed is:

1. A heat exchange unit of a refrigeration cycle device, the heat exchange unit comprising:
 - a housing having openings, and
 - gripping members attached to the openings, wherein each gripping member of the gripping members has:
 - a main body having an inner space and that opens to an outside of the housing, and
 - a plurality of claws that protrude from the main body, and that hook to a circumferential edge of the openings from an inside of the housing,
- the plurality of claws include:
 - a plurality of first claw portions, and
 - at least one second claw portion that is provided on an opposite side to the plurality of first claw portions in a predetermined direction that intersects the opening direction of an opening,
- at least one claw portion out of the plurality of first claw portions and the at least one second claw portion is an elastic claw that is cantilevered by the main body, and the elastic claw is elastically deformable in the predetermined direction with respect to the main body, wherein the plurality of first claw portions include:
 - at least one fixed claw, and
 - at least one elastic claw, and

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the protrusion height of the at least one fixed claw with respect to the main body is less than the protrusion height of the at least one elastic claw with respect to the main body.

2. The heat exchange unit according to claim 1, wherein a protrusion height of the at least one second claw portion with respect to the main body is greater than a protrusion height of the at least one elastic claw of the plurality of first claw portions with respect to the main body.
3. The heat exchange unit according to claim 1, wherein the main body has a through hole that penetrates a wall portion that configures the main body, and the at least one elastic claw of the plurality of first claw portions is cantilevered by an inner edge of the through hole.
4. The heat exchange unit according to claim 3, wherein the at least one elastic claw of the plurality of first claw portions is connected to an edge, out of inner edge of the through hole, of a side that is far from the opening.
5. The heat exchange unit according to claim 3, wherein the at least one elastic claw of the plurality of first claw portions is provided on an upper side of the main body in a vertical direction.
6. The heat exchange unit according to claim 1, wherein the fixed claw has a first facing surface that faces the circumferential edge of the opening, the at least one elastic claw of the plurality of first claw portions has a second facing surface that faces the circumferential edge of the opening, and the second facing surface is located on a side closer to the circumferential edge of the opening than the first facing surface.
7. The heat exchange unit according to claim 1, wherein the at least one elastic claw of the plurality of first claw portions comprises a plurality of elastic claws.
8. The heat exchange unit according to claim 1, wherein the at least one fixed claw of the plurality of first claw portions comprises a plurality of fixed claws, and the at least one elastic claw of the plurality of first claw portions comprises a single elastic claw.
9. The heat exchange unit according to claim 1, wherein each gripping member of the gripping members has a frame body of a frame shape along an opening edge of the main body, and the frame body is located on the outside of the housing, and sandwiches the circumferential edge of the opening in between with the plurality of claws.
10. The heat exchange unit according to claim 1, wherein the heat exchange unit is an outdoor unit.

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